The decision about a person: who looks, who signs, who pays

Lecture 7

Today, someone's scan will be read before a person ever looks at it.



Reading chest X-rays in England

2.8 million

chest X-rays read by the model in a year

out of 8.19 million taken in England in the year to March 2024

NHS England, Diagnostic Imaging Dataset Annual Statistical Release 2023/24, table 6

The care-delivery cycle: nine stages and each one's decision

While deciding

Primary screening and selection for examination
whether to call them in to look

→

Instrumental and laboratory diagnostics
what was measured

→

Diagnosis
what explains the finding

Decided and treating

Prescribing therapy and dose titration
what to treat with and how much

Surgical intervention and intraoperative monitoring
whether to operate and how

After the decision

Ongoing monitoring and clinical alerts
now or on the next round

→

Medical record-keeping
what the next reader will learn

Wrapped around every stage at once are two more: clearance, oversight, and accountability — who answers for it; professional qualification — whether the person stays in shape. **Before the cycle begins:** drug development.

The order of the first stages is held by three pillars:

- **Who was invited.** You can only look at someone who was invited.
- **Which group the numbers were computed on.** Rates are computed on the group that selection assembled.
- **The threshold for "sick."** It is set by protocol, priced against the outcome — not by the developer.

Device clearance for a model-equipped device: paths and counts

VIA THE SHORT TRACK

1,553 of 1,614

by resemblance to an already-cleared device

VIA THE FULL CYCLE

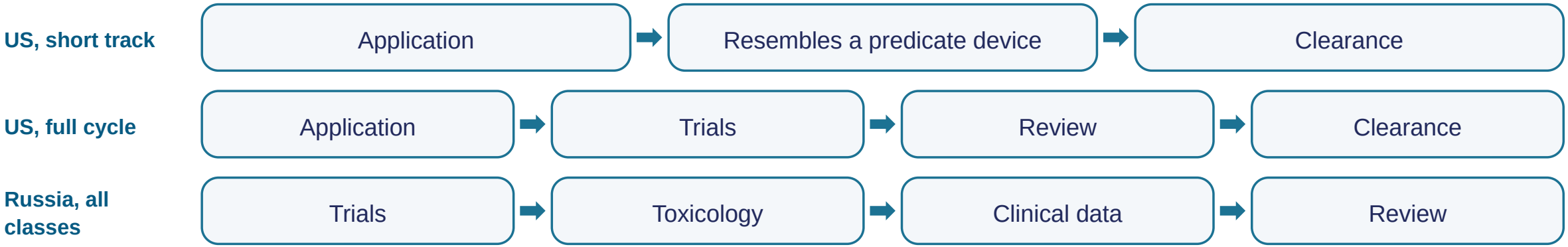
21 of 1,614

with the device's own clinical trials

REGISTRATION IN RUSSIA, ALL-IN

RUB 0.6–3 million

class 1 → class 3



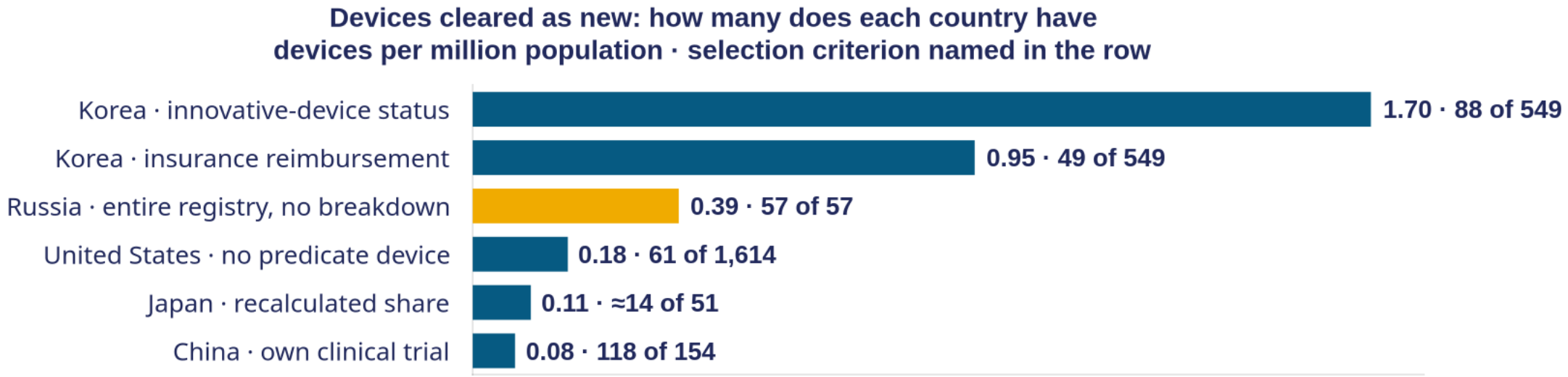
The full US track is close to the Russian one; Russia has no short track

Without a registration certificate, a device cannot be used on a person or bought with budget money.

Truly new devices are added at a rate of 3.1 a year: 56 by the end of 2025, against 7 in 2010. The rest of the growth comes through resemblance to a predecessor.

Полная запись источников в репозитории курса · допуск в США, ступени и доли · допуск в России

Devices with a model: composition of the pool and countries



RADIOLOGY

1,230 of 1,614

zero in psychiatry across all history

NEW US MEDICAL DEVICES, 2025

one in ten

in 2015 — two in a thousand

FIRST RUSSIAN DEVICES, 2020

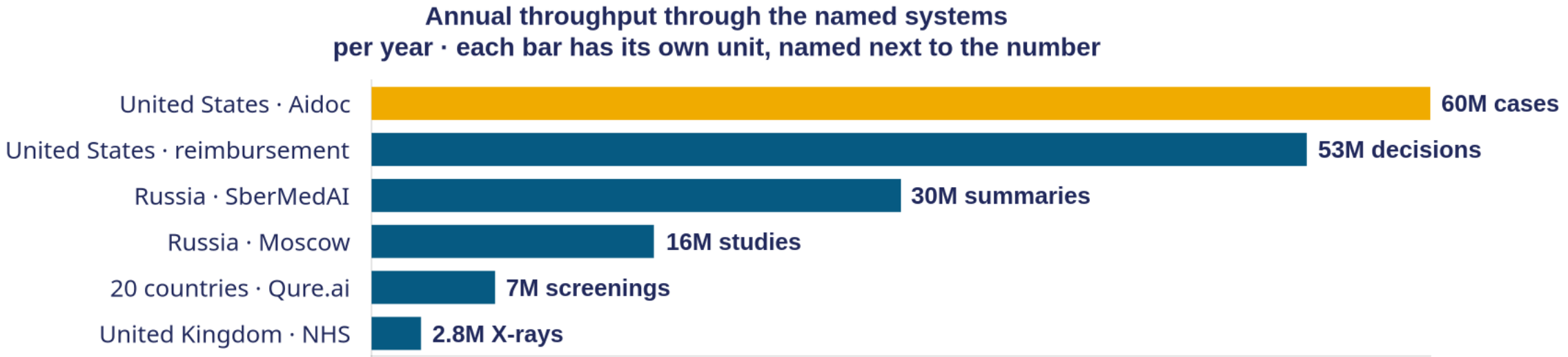
2 of 57

no predicates existed, both went through trials

The paths are counted — now the pool itself. Korea has two selection criteria: 1.70 by innovative-device status, 0.95 by reimbursement. Russia is 0.39 across the whole registry — higher than the US, Japan, and China.

Полная запись источников в репозитории курса · счёт по таблице FDA · страновой разрез · проверка Кореи и Китая

Using the model: doctors and the flow of studies



US DOCTORS

38% → 66%

of 1.1 million practicing, 2023 and 2024

CLEARED AND REIMBURSED

3 of 1,614 · 49 of 549

permanent payment codes, US · reimbursement, Korea

PEOPLE WITHIN COVERAGE

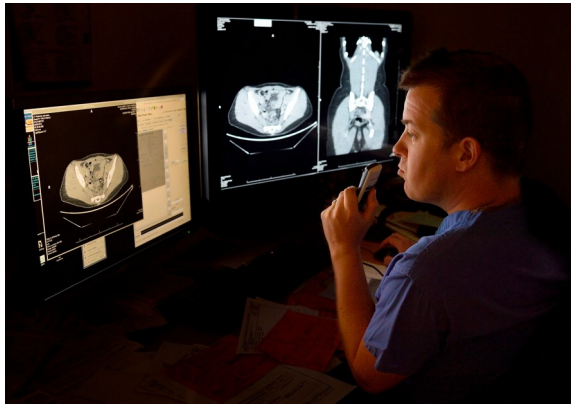
230 million

Viz.ai, 2,000 hospitals; reach, not cases

Doctors worldwide: 3 million of 13–15 million. Russia: 15–25% of 770,000. Who has a model standing in their loop — and who looks into its failures?

Полная запись источников в репозитории курса · агрегаты внедрения · использование и эффект

Six distortions at the front door: the names



Physician



Patient

- **Anchoring.** The conclusion shifts toward the model's suggestion, even when it is plainly wrong.
- **Narrowing.** Where the model stays silent, the eye does not linger.
- **Skill decay.** Once used to the model, people find less without it.
- **Alert stream.** Alerts come one after another, and so does dismissing them, one after another.
- **Disagreement.** A mismatch with the model is now handled as a separate step, and that step is new.
- **Correction.** The one that works in our favor: the doctor catches the model's mistake and fixes its report.

Полная запись источников

How decisions are made without a model, and the four rungs of human participation

human participation — down the rungs

Looks first

Looks second

Looks only at disputed cases

Signs without looking

the regulator's requirement is met on all four

The regulator's requirement: the person is the last link. The European rule puts it more precisely: they must retain the capacity to disagree and overrule. It does not require presence at every single decision.

How decisions are made without a model — the classical regimes, by name:

• **Selection** — double reading by two independents, arbitration by a third on disagreement.

• **Diagnostics** — two rates at once: how many were found among the sick and how many of the healthy weren't needlessly alarmed, both at the protocol's threshold.

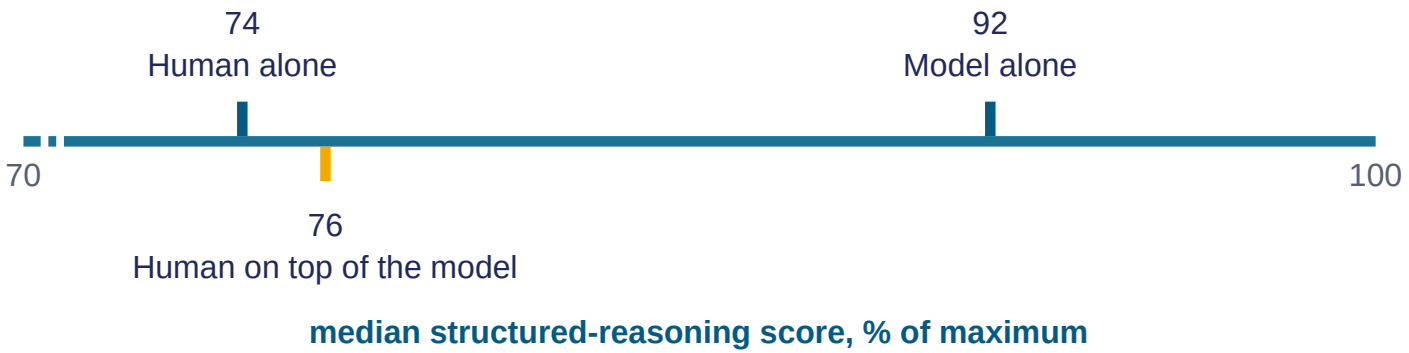
• **Diagnosis** — write out what could explain it, order the test that distinguishes between them; a second opinion, a case conference.

• **Monitoring** — an early warning score: points, a sum, a threshold for escalation.

• **Documentation** — record structure, proofreading, signature, discharge summary.

• **Oversight** — a case review after the outcome, and a change to the order of work.

Example of a human–AI system at work: 50 doctors read case write-ups





the gap between the pair and the solo human is 2 points, not significant

Three seconds and zero rubles, against 48 seconds of human reading. *The units are different — the comparison is only on order of magnitude.*

How it was measured. Fifty practicing doctors each read five real case write-ups and built out their reasoning: hypotheses, justification, which tests to order. They were scored blind, on a structured-reasoning scale. The two human conditions were randomized; "model alone" was not part of the randomization.

Reading the data got cheaper — it's paid for by a step the human does.

What pays for it: attention — looking after the model · skill — stops doing it themselves · signature — put down without reading · **arbitration — now costs more than the old reading did**. And the check of the suggestion itself has been measured: **1.99 seconds** by median.

The model looked first. What does the person do now — and who measured that?

Goh et al., JAMA Network Open 2024

1. Primary screening and selection for examination

Who to look at in the first place. Screening is examining people with no complaints, by a formal criterion. Before anyone gets treated, someone decides who's even worth looking at.

In order:

Example 1, without a model: a Swedish program calls people in by age

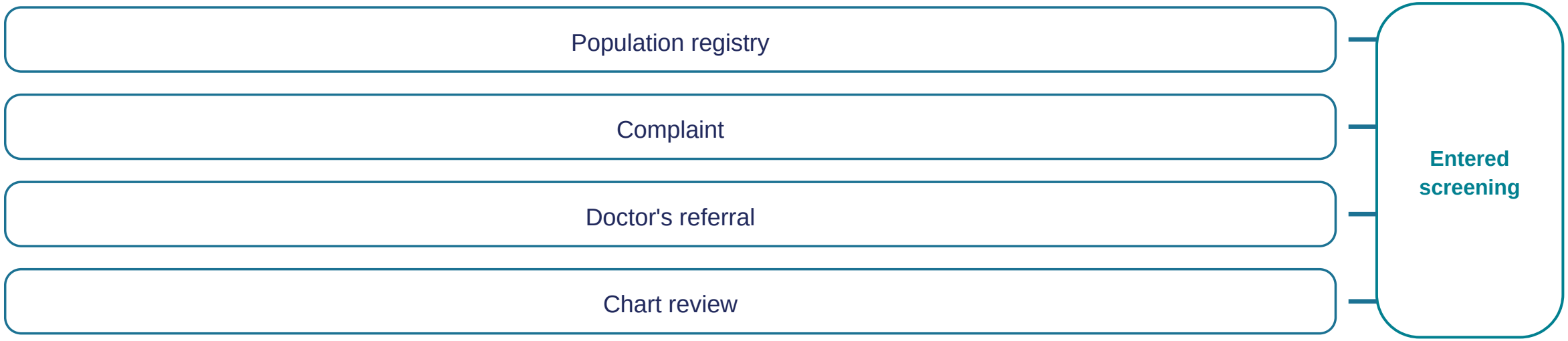
Example 2: a first-line chatbot replaces a call to the practice

Example 3: the retinal fundus is photographed at the visit

Example 4: Moscow, Russia — a model stands in as the second reader for mammograms



Primary screening: what kind of stage is this



Four entry points, none follows from another

- Known about the person: barely anything — age · sex · interval
- One question: whether to call them in to look. What follows is either an examination, or nothing

Why this stage comes first. Nobody sees the uninvited; the threshold is set by a document before any accuracy is ever measured.

Without a model, there are three orders of work: call in by age · wait for a complaint · sort those who've already come in by severity.

Seven actions of the screening stage — and three where the model switches on

Without a model

Set the threshold → Call in by list → Take the scan → Read it by two

…and further

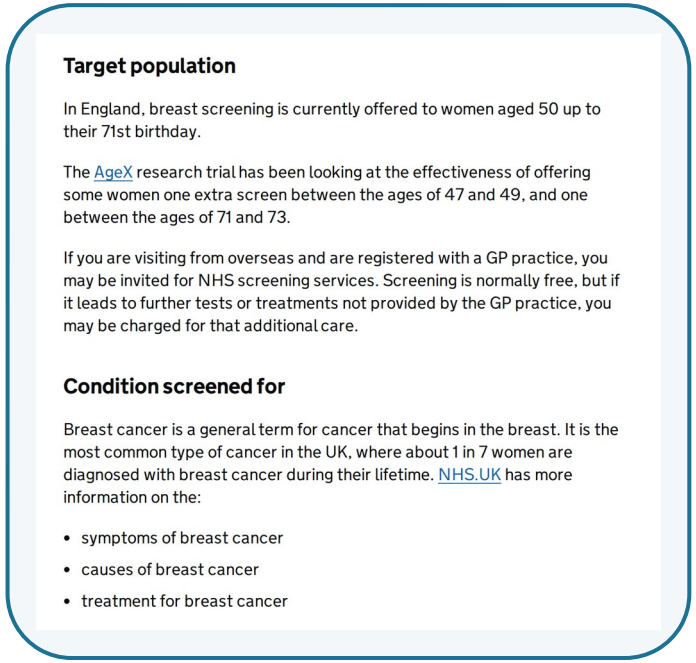
Arbitration → Report the outcome → Sort the flow

Where the model switches on

Read it by two
the Swedish program

Report the outcome
retinal fundus at the visit

Sort the flow
first-line chatbot



Target population

In England, breast screening is currently offered to women aged 50 up to their 71st birthday.

The AgeX research trial has been looking at the effectiveness of offering some women one extra screen between the ages of 47 and 49, and one between the ages of 71 and 73.

If you are visiting from overseas and are registered with a GP practice, you may be invited for NHS screening services. Screening is normally free, but if it leads to further tests or treatments not provided by the GP practice, you may be charged for that additional care.

Condition screened for

Breast cancer is a general term for cancer that begins in the breast. It is the most common type of cancer in the UK, where about 1 in 7 women are diagnosed with breast cancer during their lifetime. NHS.UK has more information on the:

- symptoms of breast cancer
- causes of breast cancer
- treatment for breast cancer

Seven actions of the stage — and three steps where the model switches on

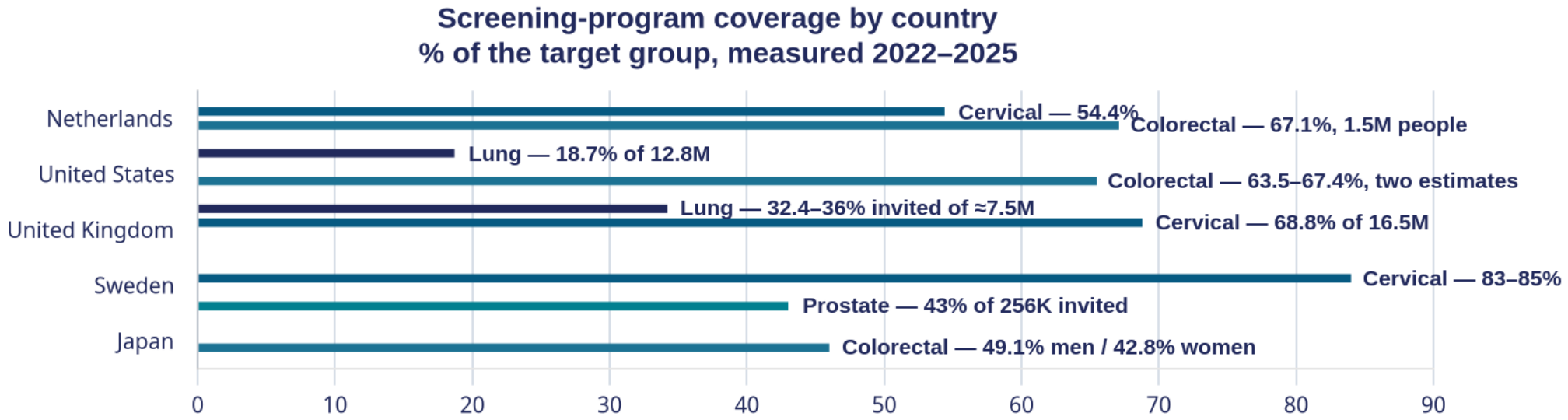
The letter the lecture opened with came out of exactly this: the program takes the list of residents, selects by age and sex, and sends out invitations. The person didn't come in on their own — they were called.

The threshold, the invitation, the scan, and arbitration stayed with people. The new step is getting a usable scan.

Whose working loop has an automatic filter in it — and who reviews what got filtered?

NHS England breast screening program, GOV.UK, accessed 13 October 2026

Screening programs by country, and where the model takes part



MODEL IN THE ARCHITECTURE

Poland · Croatia

lung, since 2020; sources give no coverage figure

MODEL WITH SEPARATE PAYMENT

Japan, since 2024

colorectal: 43% and 28% of hospitals

EFFECT ON PATIENT OUTCOME

measured nowhere

participation, coverage, and payment are measured

Row = country, column = program. No column, no program: prostate in the US, lung in the Netherlands — a decision both times, not a data gap.

National screening-program reports, 2022–2025 — ten sources are listed in the notes

Example 1, without a model: a Swedish program calls people in by age

FOUND AT THE EXAMINATION

5.1 per 1,000

Sweden, 80,033 women, double reading

MISSED BY THE PROGRAM, FOUND BY A SYMPTOM

1.76 per 1,000 over two years

control arm, 52,872 women

CALLED IN FOR NOTHING: A SECOND IMAGE

2.0 out of a hundred

repeat image, sometimes a biopsy

WAITING FOR A COMPLAINT: LATE STAGE AMONG THOSE FOUND

57 → 39 out of a hundred

Malmö, 1976–1986, invitation by randomization

Everyone is called in by age and interval. The program sends the invitation, two people read the scan, a third resolves a disagreement. There's no model here.

Waiting for a complaint is the second order of work, and it's more widespread. It doesn't work at zero: the uninvited will come in on their own, later and at a bigger stage.

The threshold is a document: age and interval are set before anyone ever measures a device's accuracy — and set by the same decision that decides whether the program exists at all.

The woman with the letter was called in by the first order: only her year of birth, none of her own traits.

Swedish program, control arm · the Malmö trial

Example 1, with a model: who gets the second reading

OUT OF A HUNDRED WOMEN WITH CANCER — FOUND

74 → 81

105,934 women, randomized into two arms

CALLED IN FOR NOTHING: A SECOND IMAGE

2.0 → 2.2 out of a hundred

80,033 women from the same program

LESS SECOND HUMAN READING

by 44.3 percent

the same 80,033 women

The same program and the same invitation. One thing changed.

What they did. Women were randomized. One arm: two readers, a third on disagreement. The other: a radiologist reads first and issues their own conclusion, while the model scores the scan for risk from 1 to 10: at 1–9 there's no second reading, at 10 double reading stays.

The conclusion is issued by the human. The model selects who gets read twice.

How it's built: input — a mammogram, output — a risk score. Publications don't give the system's name; standing in for it are Transpara, Lunit INSIGHT MMG, and iCAD ProFound. *Image recognition.*

Swedish program, final analysis

Example 1, the cost: arbitration and trust in a finding

READINGS SENT TO ARBITRATION

up two- and fivefold

two centers, 50,000 women: 3.9→9.4 and 11.3→13.8

ONE ARBITRATION REVIEW, BY TIME

equals five readings

48 seconds against 2 minutes 12; 441,276 cases

FINDING CONFIRMED: RADIOLOGIST → MODEL

3 → 22 out of a hundred

54,991 women

Without a model	Two radiologists read	Disagreement on recall	Arbitration: a third reader
With a model	**Radiologist and model read**	More disagreements	Arbitration: longer, more often

A British trial counted the cost of disagreement, the fifth of the six distortions from the front door: more disagreements, and each one is reviewed by a third reader.

Trust runs backward. A radiologist's finding gets re-checked three times as often as the model's finding, even though it's confirmed six times less often.

Germany replaced none of the human readings; 56.7 out of a hundred is a recalculation, not a measurement.

Training, updates, and drift aren't disclosed by sources.

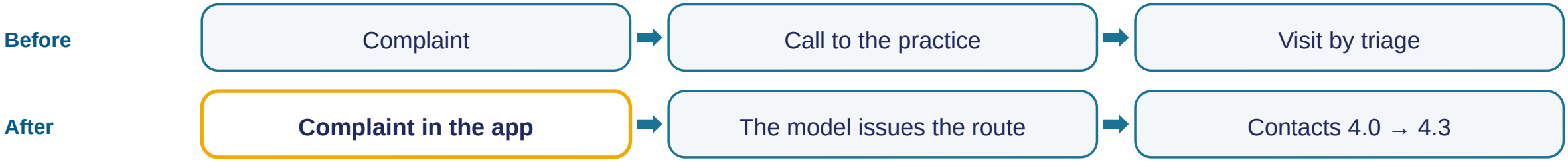
Warren et al., Nature Cancer 2026

Example 2: a first-line chatbot replaces a call to the practice

EXTRA THE COMMISSIONER PAID IN A YEAR

£10.5 million

a panel of 49,000; the whole capitation payment for it is £7.6 million



The practice is paid £155 per person a year, not tied to the number of contacts

What kind of case this is. A first-line service in England: a person describes a complaint, and the model issues the route. The second classical order of work — a phone triage line — is named, but not counted.

The cost of both errors: missing someone who needs help now · tying up an appointment · the cost of the channel.

Exam: 81 for the app against 72 for doctors. This is what justified the move into the open flow; nobody measured the quality of the triage itself.

What it gave: round-the-clock triage · **what it cost:** a loss on every patient · **net result: worse off** — bankruptcy in 2023.

Royal College of GPs, 2018 · independent service evaluation, 2019

Example 2, the architecture: who said what about what's inside Babylon

THE DECISION OUTPUT IS ONE OF

6 options

from 'call an ambulance' to 'you can handle it yourself'

The ordinary path	Symptoms as input	**The graph computes — the neural net makes it faster**	One of the outcomes as output
A red flag	Symptoms as input	A rule — bypassing the graph	Always: an ambulance

A red flag bypasses the graph entirely

Three people spoke, and that's not one source. Hugh Harvey, the company's head of regulatory affairs in 2016–2017: inside, a decision tree of "if-then" rules in a table. **David Watkins**, a doctor outside the company: he's been writing to the regulator about missed cases since 2017. **The company itself**, a 2016 publication: a probabilistic graph plus hard-coded rules for red flags on top of it — on the diagram.

Words from the inside · letters from the outside · the company's own publication. Not one independent check exists.

A trained model making the decision was never here at all.

Middleton et al., arXiv:1606.02041, 2016 — a publication by the company itself

Example 3: the fundus is photographed at the appointment

STEP 1 — REACHED THE EXAM

22 → 100 out of 100

81 versus 83, by randomization

STEP 2 — REACHED THE OPHTHALMOLOGIST

16 people out of 25

that is 64 out of 100; in the control, the same 22

Without the model	Referral	Waiting for the visit	Specialist reading
With the model	Image taken by the nurse	Autonomous reading	On-the-spot decision

What this case is. Adolescents aged 8–21 with diabetes: once a year they are checked for a complication — fundus damage. This is a check of an at-risk group, not screening of healthy people.

The cost of both errors: a miss — asymptomatic until vision loss · a false call — a visit to a specialist who may not exist in the area · and they lose people to no-shows.

Both classical regimes are counted in attendance: to the ophthalmologist once a year — 22 out of 100 · not screening at all — zero.

Wolf and co-authors, Nature Communications 2024

Example 3, cost: the reject stream and a new step

UNGRADABLE FOR QUALITY

0.9 out of 100

910 of 101,710 visits, 404 clinics, adults

STREAM THE SPECIALIST NEVER SEES AT ALL

75.7 out of 100

previously, normal cases reached him and he could check them

Before — The ophthalmologist reads it → The same person answers for it

Now — **The device reads it** → Responsibility rests with the device → New step: a gradable image

Imaging at the appointment delivered attendance. Now — what it cost, and to whom.

The performer changed on paper — the only such case in this lecture: responsibility for a missed finding moved from the physician to the device.

The ophthalmologist never had this new step: he adjusted the imaging conditions himself.

Both figures come from a different program and a different device, in adults.

delivered on-the-spot access · **required** a new step and a stream nobody reviews · **net**: access bought at the cost of a reject stream.

US regulator decision DEN180001, 2018

Example 3, mechanics: training, updates, and a different population

UNGRADABLE IMAGES AT THE HOSPITAL

26 out of 100

875 patients, same device, different population

DETECTS: SEVERE → MODERATE FORM

94 → 61 out of 100

at authorization, the same device gave 87 and 91

IMAGE DEFECT RATE: DAY 1 → DAY 4

47 → 0 out of 100

the same operator throughout

Before authorization

≈25,000 exams → **Model training** → Regulator authorization, 2018

After authorization

Different population → Detects worse, rejects more → Protocol update

How it's built: input — two fundus images; output — one of three outcomes: refer to an ophthalmologist · normal, repeat in a year · image is ungradable. IDx-DR (now LumineticsCore) was the first authorized, in 2018. *Image classification.*

Authorization does not freeze the device: the quality classifier was updated through a submission.

These are two different effects: disease severity versus skill at holding the camera.

German study of IDx-DR, 2026 · submission K213037

Example 4: Moscow, Russia — a model standing in for the second mammogram reader

SCALE OF THE CHECK

593,261 studies

112 Moscow organizations, 10 equipment manufacturers, 2021–2022

SENSITIVITY: ENTRY → EXIT

0.62 → 0.85

that is 38 cancer signs out of 100 not found

REGISTRATION CAME BEFORE THE CHECK

2021 — Roszdravnadzor certificate and CE mark

for the version with 0.62 sensitivity

Double reading	image	first physician	second physician	a third on disagreement
Moscow, with the model	image	**the model reads first**	the physician reads second	the physician's finding

The model took the second reader's slot, not a spot next to the first

What this case is. Moscow, mammography screening. The model reads the image first, the physician second. *The product is "Celsus Mammography."*

How it's built: input — a mammogram in four views, output — a number from 0 to 1 for each breast. Faster R-CNN with a ResNet-50 backbone, trained on 3,641 images. *Image recognition.*

There is no outcome benefit: detection rate, recall rate, and interval cancers were not collected at all — the authors say so themselves.

Vasilyev, Rumyantsev, Vladzimirsky et al., Diagnostics 2025

Example 4, authorization: two years and nine months to insurance payment

TECHNICAL FAILURES

9 → 1 out of 100

most are not the model: device setup, technician parameters, archive failures

WHAT ACCURACY WAS MEASURED AGAINST

100 studies, not 593,261

the benchmark is the finding of the very physician the model replaces: it cannot be measured above him

THE PHYSICIAN DISAGREED WITH THE MODEL

1,745 times out of 8,406

33 radiologists; this is "Disagreement" — the fifth mechanism from the opening slide

Check	output requirements	calibration on 100	stream and failures	review by an expert physician
Response	report to the developer	**new model version**	re-calibration	insurance payment

The review cycle: a report goes to the developer every month

The first two objections were not about accuracy: the finding was highlighted in color, and mammography monitors are monochrome.

Area under the curve: 0.73 → 0.81 → 0.94. *Scale: 0.5 is a coin flip, 1.0 is no misses. The 0.81 threshold is the lower bound of the radiologist's own confidence interval.*

Vasilyev, Rumyantsev, Vladzimirsky et al., Diagnostics 2025

Section conclusion: a gain in access, no added accuracy from the model

What works: reaching people who otherwise would not come — **22 → 100 out of 100** · cutting the volume of reading where the procedure is already dense — **44.3 percent less second reading**.

What does not work: setting the threshold — a signature stands behind it · an open-referral stream — the one attempt that actually happened ended in bankruptcy · answering for the gradability of the image.

With the work: the workload moved from triage to the visit, and there is now nobody to pay for it · the performer changed, and a step appeared — obtaining a gradable image.

No procedure checks what the filter screened out. In screening, a human read every study; where the model decided, its "normal" never reached him.

Of the six distortions named at the opening, this section ran into one — disagreement: arbitration grew two- and five-fold, and one review takes as long as five readings.

Nine forms of the move: what each looks like in an example

	form	what it looks like in the example
1	the workload moved to a different step	triage moved to the visit — nobody to pay for it
2	the step disappeared entirely	the candidate was checked and dropped out
3	a step appeared	the decision of who needs a second reading
4	the workload grew in the same place	alarms are frequent, confirmation is rare
4a	the workload in the same place fell	the same stream, less work
5	the performer changed	the device issues "refer or not"
6	the order of work changed	rearranged, but not counted
7	the cost of an error shifted	the model's error is the signer's error
8	the performer is the same, but got worse	without the assistant, they find less
8a	*the performer is the same, but got better*	*no measurements exist*

2. Instrumental and Laboratory Diagnostics

In screening, the decision was who to examine. Here the person is measured by a device, and their condition is turned into a value that will be read somewhere else, at another time.

In order:

Example 1: Moscow — preventive scans that no human reads

Example 2: Moscow — intracranial hemorrhage, where autonomy was withheld

Example 3: Thailand — fundus reading in a primary-care office

Example 4: skin, and the composition of the dataset

Example 5: selection into a high-attention program — a score instead of a physician's decision



The evidence: six actions of the stage, and the two the model took over

Human — four

set the threshold → take the measurement → decide if it's usable → recompute for its own group

Model — two

read the measurement

report the result

Six actions of the stage; the model took over two



Four ways to take a measurement, and none can substitute for another:

- **Instrumental examination** — the device works directly on the person
- **Laboratory analysis** — the person is no longer present, a retake on the spot is impossible
- **Functional test** — a load protocol is part of the measurement conditions
- **Remote examination** — the reader never saw the patient

The model took over two of the six actions, and not the hardest two. It does not set the threshold against the cost of an error, cannot order a retake, and leaves input gradability to a human.

You're told "94 percent accuracy." On what group was it computed?

Roberts et al., Nature Machine Intelligence 2021

How accuracy is counted: one trigger threshold on two groups of ten thousand

FINDS IN 100 SICK PEOPLE · FALSELY ALARMS IN 100 HEALTHY PEOPLE

94 · 11

one published setting, applied to both groups unchanged

OUT OF 100 ALARMS TURNED OUT TO BE DISEASE: STREAM → DEPARTMENT

8 → 79

94 of 1,183 versus 2,820 of 3,590; two groups of 10,000 people each

General population stream	1 sick out of 100	1,183 alarms	8 correct out of 100
Specialized department	30 sick out of 100	3,590 alarms	79 correct out of 100

One system, one trigger threshold; what differs is the share of sick people in the group

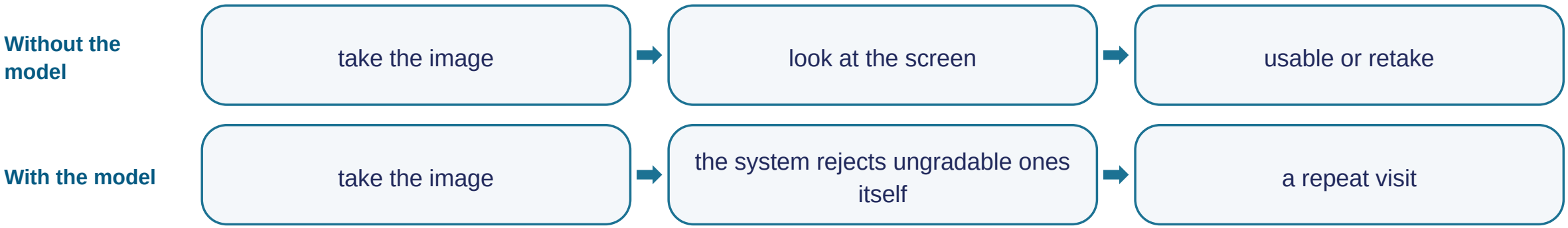
The properties of a measurement are computed on a group. One **trigger threshold** — the value beyond which the system declares "sick" — was applied to two groups of ten thousand people, differing only in the share of sick people.

The difference is nearly tenfold, and not one line changed inside the system. This share has a name — **positive predictive value**: what fraction of alarms turned out to be disease. This is what the physician at the visit actually sees, not sensitivity.

The woman with the letter stands in the general-population stream: for her it is 8 out of 100.

Rajpurkar et al., CheXNet 2017

The classical regime: image gradability is decided by the technician at the machine



The decision on image gradability: without the model and with the model

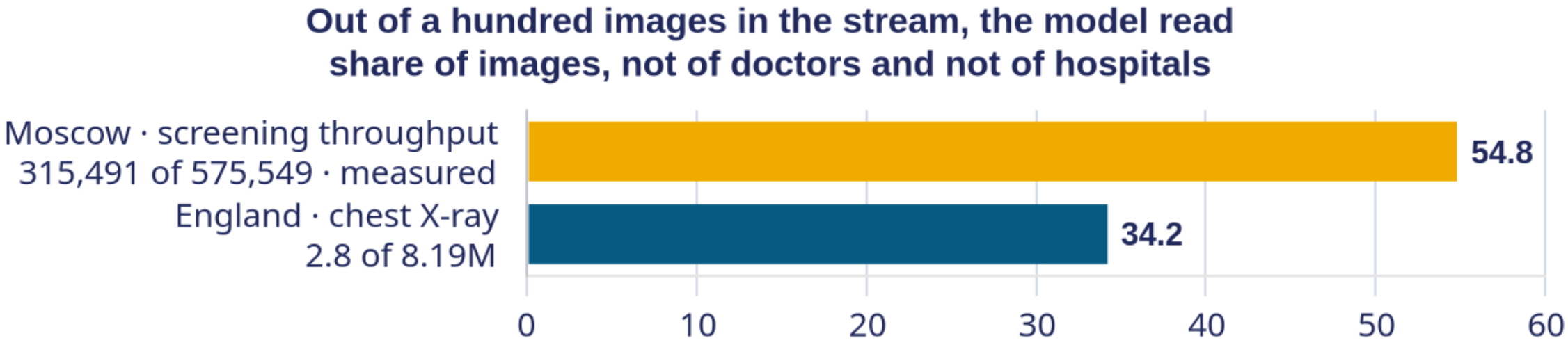
The classical regime of this stage — **quality control of the measurement.**

The rule the technician decides by: the room's own instructions and the visible quality of the frame. The device carries no such rule.

What happens if an ungradable image goes to reading anyway: the finding gets made on what is visible — and down the chain follow a wrong diagnosis, a repeat study, lost weeks, and treatment started too late. **Nobody notices,** until a review is needed.

With the model, the fork does not disappear — it splits in two: it rejects an ungradable image itself — and then the person gets a repeat visit; if it does not reject it, it reads whatever it was given.

How much of the stream the model reads: one basis for three countries



Gradability of every image was decided by a human at the machine — now about the stream itself. There is one question: out of a hundred chest images taken in a country, how many did the model read. Any other comparison of countries here is a splicing of samples.

The English share has two official denominators. The numerator, 2.8 million, is stated by the developer. Health-service statistics give 8.19 million X-rays — which works out to 34 out of 100; the government's own announcement names "over 7 million" — the same 2.8 million gives 40 out of 100.

The US has no figure of this kind at all. What gets published is the share of products in the registry (76.5 of 1,430 over thirty years) and the share of physicians using the model (34 by survey). The second looks like the English figure down to matching digits, and measures something else.

Vasilyev et al., Digital Diagnostics 2025;6(1):5–22 · *NHS England, Diagnostic Imaging Dataset 2023/24* · *UK government announcement*

Moscow, the preventive stream: how many images no human opens

55 out of 100

preventive chest images that no physician opens

315,491 of 575,549 over five months of 2024, Moscow

the model's finding goes straight into the electronic chart

Vasilyev et al., Digital Diagnostics 2025;6(1):5–22

Example 1: Moscow — preventive images that no human reads

THE COST OF BOTH ERRORS

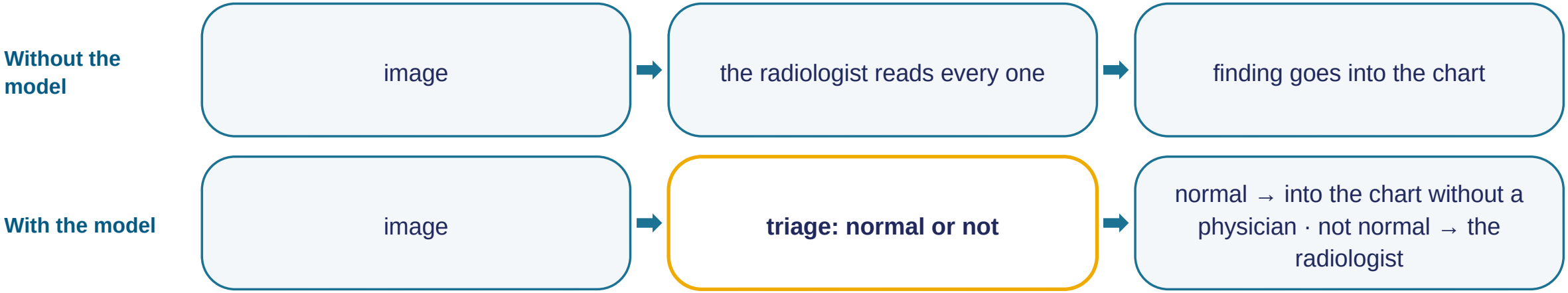
the disease is found a year later · an unnecessary follow-up exam

the asymmetry is extreme — the trigger threshold was derived from it

CLINICALLY SIGNIFICANT MISSES

5 per 10,000 images

290 of 575,549 is 0.05 percent; a human on the same kind of image gets 3–4 percent, but on a different stream



How this 55 out of 100 came about. Chest images, Moscow, May–September 2024: 575,549 fluorography and X-ray scans, three services.

How it's built. Sensitivity was set to **1.0**, specificity was declared **not significant**. 45 images out of 100 go into "not normal" — the radiologist reads these. The rest go into the chart without him.

What was missed: infiltrate or opacity — 266 of 290 cases.

Vasilyev et al., Digital Diagnostics 2025;6(1):5–22

Example 1, cost: it got cheaper only because nobody reads half of it

HUMAN READING OF ONE IMAGE: BEFORE → NOW

132.98 → 155.62 RUB

got more expensive on every image still read by a human; the model's finding — 6.9 RUB

OVER FIVE MONTHS: WITHOUT THE MODEL → WITH THE MODEL

76.5 → 43.1 million RUB

−43.7 percent, savings of 33.4 million RUB under mandatory health-insurance tariffs

HALF THE STREAM IS PEOPLE AGED 18–44, AND THEY HAVE MORE MISSES

12 versus 7 per 10,000

155,898 of 315,491 "normals"

Where the savings come from

45 out of 100 is read by a physician — pricier than before → 55 out of 100 is read by nobody → net −43.7%

What had to be done beforehand

amending the mandatory health-insurance tariff agreement → four new service codes → without it, there is nothing to pay for the model's finding with

The benefit is not that reading got cheaper. One image read by a human got **more expensive**: 155.62 RUB against 132.98. The savings come entirely from those 55 images out of 100 that nobody reads.

The layer that found the 290 misses will not exist in routine operation: all the "normals" were re-read by humans under the study's own protocol. The share will remain; the way to see it will not.

Delivered: −33.4 million RUB over five months. **Required:** 290 missed findings, and amending the tariff agreement before the work could even start.

Vasilyev et al., Digital Diagnostics 2025;6(1):5–22

Example 2: Moscow — intracranial hemorrhage, where autonomy was withheld

THE COST OF BOTH ERRORS

death or irreversible damage · a crew deployed for nothing

it comes down to minutes — there is no time to check an alarm

FALSE ALARMS ON 3,409 SCANS: MODEL ALONE → PHYSICIAN

293 → 4

73 times more; every twelfth scan would have gone down the emergency path for nothing

THE MISSES NEVER OVERLAPPED ONCE

45 and 12

the physician found all 45 the model missed; the model found all 12 the physician missed

What was proposed	scan	**the model reads alone**	the emergency path
What the measurement showed	scan	**the model flags a suspicion**	the physician decides — and stays the first reader

The same Moscow, the same circle of services — the opposite answer. 3,409 head CT scans, 67 clinics, 2022–2024; hemorrhage in 1,101 people.

This reads by count, not by share. Out of 100 sick people, the model finds 96, the physician 99. False alarms: **293 against 4**.

Caveat. Here the physician worked **with the model's cue**; there is no measurement of "the physician without the cue" in this study.

Khoruzhaya et al., Journal of Clinical Medicine 2025;14(16):5700

Example 3: Thailand — fundus reading in a primary-care office

THE COST OF BOTH ERRORS

vision loss · a visit to an ophthalmologist who does not exist

no monetary estimates exist for this country

IMAGE REJECTED FOR QUALITY → A REPEAT VISIT

21 out of 100

393 of 1,838 over the first six months

Without the model	referral to a specialist	waiting for the visit	specialist reading
With the model	image taken by a nurse/technician	**autonomous reading**	refer / negative / rejected for quality

Fundus reading in a primary-care office. Six months, eleven clinics in Thailand: nurses and technicians took the images, the system read them on the spot in under a minute and rejected unusable ones itself.

How it's built. Trained on images already selected for quality; in the field, that selection is made by a human with no rule. The system's name is not given. **Reasons for rejection:** lighting, a slow connection.

The deployment failed, the reading worked. *A field observation, measured once; orders of magnitude are compared, not samples.*

Beede et al., CHI 2020

Example 4: skin, and the composition of the dataset

THE COST OF BOTH ERRORS

treatment at a late stage · biopsy and a scar

OUT OF 100 SICK PEOPLE FOUND: NAKED EYE → LIT MAGNIFIER

86 → 91

out of 100 healthy people falsely alarmed, 23 → 26; one meta-analysis

HOW MUCH WORSE THE MODEL DISTINGUISHES SICK FROM HEALTHY ON DARK SKIN

by 9–15 out of 100

scale: 50 is a coin flip, 100 is no misses. Three models, 656 confirmed-diagnosis images: 208 light · 241 medium · 207 dark

Without the model

naked-eye exam / dermoscopy ➡ decision: see a physician or not

With the model

image taken by phone/camera ➡ **the model** ➡ show to a physician or not

What was done before the model. A person finds a lesion on their skin and wants to know whether to see a physician. A physician looked with the naked eye; a trained specialist, through a dermoscope.

How it's built. Three models on one validation set, light tones against dark. Each of the three has its own training set, and dark tones are scarce in them — that is the whole matter. *Image classification.*

The classical regimes are from one head-to-head meta-analysis.

Daneshjou et al., Science Advances 2022 · *Vestergaard et al., British Journal of Dermatology 2008*

Example 4, cost: dataset composition is what heals the gap

FINE-TUNED ON A SET WITH AN EQUAL SHARE OF DARK TONES

light 73 · dark 76 out of 100

the gap didn't disappear, it flipped; what heals it is collection and labeling, not the algorithm

THREE ALGORITHMIC BIAS-CORRECTION METHODS

did not shrink the gap

same study, same data

FOR THE CLASSICAL METHODS, THE GAP

was never measured at all

neither the meta-analysis nor the reviews break accuracy down by skin tone

DELIVERED → REQUIRED

an assessment where there is no dermatologist → a dataset matching the composition of those it works on

net: the average holds up, the cost lands on the subgroup

Algorithmic correction	algorithmic correction	➡ the gap did not shrink
Dataset composition	fine-tuning on a representative set	➡ the gap closed

Of two paths against bias, one worked, and it isn't a matter of picking a method

What moved. The cost of an error shifted: the dataset average stayed respectable, while the subgroup underrepresented in the training set pays.

The absence of a measurement does not prove equality.

The fix is measured in this same study; in a real clinical setting, none of the three is.

Daneshjou et al., Science Advances 2022

Example 5: selection into a high-attention program — a score instead of a physician's decision

THE COST OF BOTH ERRORS

an unwatched illness · an occupied slot

coordinator time — someone else didn't get in

SHARE OF BLACK PATIENTS AMONG THOSE SELECTED: PHYSICIAN DECIDED → THE SCORE DECIDED

19 → 17 out of 100

49,618 patients of one hospital, 2013–2015; one sample, both regimes on it

AT THE SAME SCORE, BLACK PATIENTS HAVE MORE CHRONIC CONDITIONS THAN WHITE PATIENTS

by 26 out of 100

the score called equally healthy people whose health actually differs; 100,009 patient-years

IF THE SCORE HAD TARGETED ILLNESS RATHER THAN SPENDING: THE SAME SHARE

18 → 47 out of 100

the authors' own calculation on the same data (17.7 → 46.5); this is a computation, not a deployment — no such rollout ever happened

Physician (saw the score)	patient chart	cue: the score	the physician's decision
The score alone	chart + spending	**spending forecast**	threshold-based selection

Selection into a high-attention program. A physician used to decide — now a single number decides. **It is trained to predict future treatment spending**, while the decision being made was about illness: spending reflects utilization, and utilization differs across groups with the same underlying health.

How it's built. *Forecasting on structured data.*

The cost of an error moved — onto the group that, at the same score, has more illness. **The study did not measure the effect on money or health.**

Obermeyer et al., Science 2019

Section conclusion: the reading itself failed once out of five

THE SETTING, THE DATASET, AND THE TARGET FAILED

3 of 5

Thailand · skin · program selection

THE READING FAILED AS A READING

1 of 5

hemorrhage: 293 false alarms against 4 on 3,409 scans

AUTONOMY WAS GRANTED AND WITHHELD IN THE SAME CITY

decided by the cost of an alarm

preventive stream — a follow-up exam · emergency path — a crew

What works

- consistency: the ten-thousandth image reads like the first
- throughput: "normal" does not occupy the specialist
- reproducibility: a discrepancy can be worked through

What does not work

- setting the trigger threshold: a cost of both errors stands behind it
- ordering a retake: measurement gradability stays with the human
- seeing that the input itself is biased: no procedure can do this

You can go wrong in three places without the model itself being wrong: a device measures the input, a human assembles the dataset, a variable sets the target.

Autonomy is decided by the cost of a false alarm, not the quality of the model.

Where the model stands. *Image classification* — fundus, skin lesion. *Image recognition* — triaging the stream on X-ray and CT. *Forecasting on structured data* — a risk score in the chart. *Measurement · device signal* — **no model, and still bias.**

Of the six distortions named at the opening, this section ran into none: the stream is read by the model alone.

Five questions asked before the question of accuracy

QUESTIONS BEFORE THE QUESTION OF ACCURACY

five

each one paid for by a case already broken down

SANK A DEPLOYMENT

3 of 5

Thailand · skin · program selection

THE FOURTH SANK THE MEASUREMENT ITSELF, THE FIFTH SETTLED THE QUESTION OF AUTONOMY

the device · the cost of an alarm

the cost of an alarm is what told the two Moscow cases apart

"What's its accuracy" is question six. Five come before it.

• **On what group were the shares computed?** The set was assembled from images already selected for quality; in the field, that selection is made by a human with no rule — *Thailand*.

• **What measured the input?** The device read normal while the blood test showed oxygen deprivation: **12 times out of 100 against 4**. There is no trained model in the device.

• **What was the training set made from?** Dark tones were scarce in the sets: on dark skin, the model distinguishes sick from healthy **9–15 out of 100 worse** (100 is no misses, 50 is a coin flip) — *skin*.

• **What does the system treat as its target?** Trained on future spending, while the decision being made was about illness — *program selection*.

• **What does a false alarm cost, and is there time to check it?** One city: on the preventive stream, autonomy was **granted**, on emergency scans it was **withheld** — **293 false alarms against 4** from the physician.

Three sank a deployment each. The fourth sank the measurement itself. The fifth decides whether a human can be removed at all.

Sjoding et al., New England Journal of Medicine 2020 · US regulator warning, February 19, 2021, draft guidance January 6, 2025

3. Diagnosis

There, the stage produced values — here, a finding is made from them, and a human signs it. He now reads not only the image, but also the cue next to it: the order in which they are presented affects the outcome.

In order:

Example 1: eye-tracking — with a false cue and with a correct one

Example 2: pathologists read slides — with a cue and without

Example 3: the Swedish trial — the second human reading replaced by a model

Example 4: surgeons assess risk — on their own, and after seeing the model

and, carried over from continuous monitoring, **Example 5: an early-warning model for sepsis**

Example 6: Moscow, Russia — a model proposes a final diagnosis from the medical chart



Diagnosis: the stage's seven steps, and the three where the model operates

FREQUENCY OF DIAGNOSTIC ERROR: ADULT OUTPATIENT VISITS

5.08 out of a hundred visits

identified via a trigger — an unusual patient return, an abnormal result with no follow-up workup; so only part of the errors is counted; the share itself was extrapolated to the US adult population; the authors call this an underestimate: the true frequency may be noticeably higher than five percent



The workflow

gather the data → list candidate diseases → order a differentiating test → cross out what's ruled out

Occupied by the model

composing the list | the second reading | the sign-off

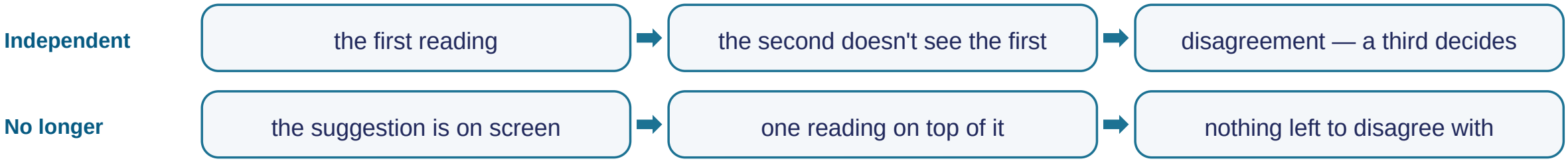
How the list narrows: the stage right after the workup

One visit: a cough for three weeks, an evening fever, a four-kilogram weight loss. Of four candidate diseases, the X-ray and blood test ruled out two — two remained, indistinguishable on the image.

The model occupies three of the seven steps: composing the list, the second reading, the sign-off. **And it added one — deciding whether to trust the suggestion.**

Singh et al., BMJ Quality & Safety 2014

A second opinion: what holds it up, and when it stops being second



Two opinions: when they're independent, and when they no longer are

Held up by independence: the second reader is denied access to the first report, and a third reader resolves any disagreement.

A second opinion from the model that suggested the first one is no longer a second opinion: the error is shared, and it doubles instead of averaging out.

The order of presentation has been measured. A suggestion shown before the person formed their own opinion, versus a suggestion shown after — **8.9 points** in decisions to act. *70 physicians, 254 cases.*

Which of you has asked a model for a second opinion on a decision you'd already made yourself? What did you do when it disagreed?

Goh et al., a continuation of the same trial

Four rungs of involvement

human involvement — down the rungs

Looks first
sees the case

Looks second
sees the case and the finished conclusion

Looks only at disputed cases
sees the sample the model selected

Signs without looking
sees only the text

The second rung holds both the gain and three failures. The eye-tracking study, the pathologists, the surgeons — and the Swedish trial, where the person stays the first reader and the model becomes the second: one rung, opposite outcomes. What differs between them is **the design of the procedure** — whether the independence of the two opinions is preserved, and who resolves a disagreement.

Training doesn't move you up a rung: 44 trained physicians read six clinical cases each — the share of correct reasoning across 264 readings **84.9 → 73.3 out of a hundred** when some of the recommendations are wrong.

The risk of the third rung: the person doesn't know what the model screened out. **The fourth** — the formal requirement that "a person checked it" is met in a median of **1.99 seconds**: not enough time to read the suggestion.

Qazi I. A. et al., NEJM AI 2026 · *Mozannar et al., CHI 2024*

Example 1: eye-tracking — with a false suggestion and with a correct one

MISSED CANCERS WITH A FALSE SUGGESTION: WITHOUT → WITH

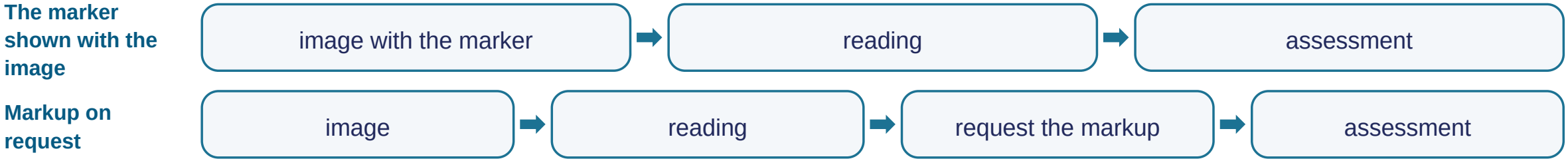
29 → 61 out of a hundred present

14 cases where the suggestion was silent; significant. The cost is later, worse treatment

MISSED CANCERS WHEN READERS REQUESTED THE MARKUP THEMSELVES: WITHOUT → WITH

165 → 115 out of 900 readings

75 cancer cases per condition; a 5.6-point gain with an interval of −0.3… +11.4 — not significant. Reading time didn't change



The two studies diverge by the order in which the suggestion was presented

Two measurements of one quantity. 10 radiologists read 60 mammograms twice: on their own, and with a suggestion deliberately wrong in both directions. 12 radiologists in a different program: 150 exams, also read twice, with an approved system.

A different order of presentation — the opposite sign.

How it's built. Input — a mammogram, output — a suspicion score for the region. The second study names the product: **Transpara 2.1.0, ScreenPoint Medical. Focus:** the area of the image scanned fell from 11.1 to 9.5 out of a hundred, with longer dwell time in the right spot — 4.4 → 5.4 seconds.

This narrowing is the second of the six distortions from the opening of the evening: where the model stays silent, the gaze doesn't linger, and the model's miss becomes the reader's miss.

Taib et al., Radiology 2026 · *Gommers et al., Radiology 2025*

Example 2: pathologists read slides — with a suggestion and without

SHARE OF CORRECT FINDINGS: MODEL ALONE · PATHOLOGIST ALONE · THE PAIR

0.842 · 0.898 · 0.914

intervals of 0.875–0.916 for the pathologist and 0.893–0.930 for the pair; the pair's gain over the pathologist at the group level is NOT robust — odds ratio 1.281. The model's accuracy is on its own independent test set

THE SAME PAIR BROKEN OUT BY WHETHER THE SUGGESTION WAS RIGHT: CORRECT → INCORRECT

≈0.94 → ≈0.50

converted from odds ratios of 4.289 and 0.253, both robust, from a baseline of eight correct out of ten: the study publishes odds, the shares here are illustrative

IN SUM

the average is the same, the spread grew

Without the model

slide on screen → pathologist decides → sign-off

With the model

slide on screen → **the model's suggestion** → pathologist decides → sign-off

Half the cases with a suggestion, half without; the sign-off is the pathologist's either way

11 pathologists of varying experience distinguished two types of liver cancer across 80 digitized slides. **The cost of error cuts both ways:** mistaking one tumor type for the other means different treatment; re-checking means a specialist's time.

Here the pathologist is stronger than the model, and the pair beats both sides. At the same time, the model's error becomes the physician's error: the sign-off is theirs.

"No effect" in a one-number report: the group-level gain isn't significant, even though the effect runs fourfold in both directions.

Kiani et al., npj Digital Medicine 2020

Example 3: the Swedish trial — the second human reading replaced by a model

MISSED CANCERS: DOUBLE READING → ONE READER WITH THE MODEL AND AN ARBITER

26 → 20 out of a hundred present

randomized trial, 105,934 women; the difference is robust

INTERVAL CANCERS PER 1,000 SCREENED: DOUBLE READING → WITH THE MODEL

1.76 → 1.55

93 out of 52,872 versus 82 out of 53,043. Improvement is NOT proven: ratio 0.88 with an interval of 0.65–1.18 — non-inferiority

VOLUME OF HUMAN READING

down by 44.3 percent

baseline — double reading, two human readings per image; interim analysis, 80,033 screened

Double reading

two radiologists separately → disagreement reconciled → decision

Second reading — the model

radiologist first → **model second** → disagreement — a third reader → decision

Only disagreements go to the arbiter, not the whole stream

105,934 women in the Swedish program were randomized between two reading orders. The second order uses less human reading: only a disagreement goes to the arbiter.

This trial never measured the model alone, and there's no single reading without arbitration either: the comparison was against a strong classical mode.

Disagreement, the fifth input distortion, became a new piece of work; recalls became neither more nor fewer.

The MASAI final analysis, The Lancet 2026 — (25)02464-X · safety analysis, Lancet Oncology 2023 · Dembrower et al., Radiology 2025

An EKG with a machine readout: what happens when the readout is wrong

How they read EKG with the readout → trainee reads → report

The readout arrives before the trainee's own decision; the study never measured reading without it

30 trainees each read a hundred EKGs, each one three times: without a readout · with a single machine readout, half of them wrong · with several candidate diagnoses. 15 per group — cardiology trainees and trainees from other specialties.

With a CORRECT readout, both groups read almost identically: 84.86 out of a hundred for the cardiology group, 86.38 for the other.

Question: what happens to these two numbers when the readout is wrong — and whose drop will be deeper?

Bond et al., Journal of Electrocardiology 2018

The reveal: both groups fall by half, the less expert one falls deeper

ACCURACY LOST ON AN INCORRECT MACHINE READOUT: CARDIOLOGY TRAINEES → OTHER SPECIALTIES

−43.20 → −58.95 points

the difference is robust in both groups; the less expert group loses 15.75 points more — the authors' own conclusion, stated twice

AGREED WITH THE INCORRECT READOUT

30.77 versus 42.31 out of a hundred

medians for the same two groups, in the same order: the less expert group agrees more often

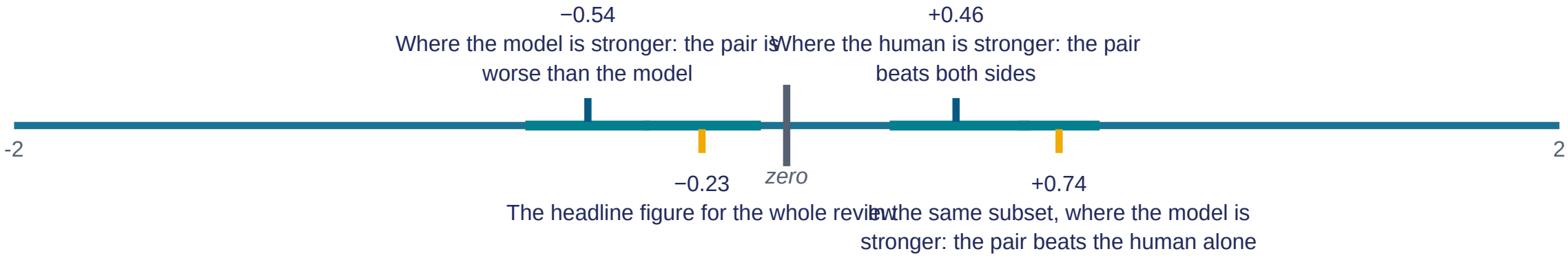
The ones who fell deeper were trainees from other specialties — those with less training in this kind of reading.

report accuracy, out of a hundred	correct	incorrect
cardiology trainees	84.86	**41.66**
other specialties	86.38	**27.43**

Anchoring, the first of the six input distortions. Specialty protects, but it doesn't save you: both groups fall, and both by half.

Bond et al., Journal of Electrocardiology 2018 · Buçinca et al., CSCW 2021 · Yu et al., Nature Medicine 2024

A review of 106 studies: where the pair beats the human, and where it falls short of the model



effect size, in units of spread · 74 studies · 106 experiments · 370 effect sizes out of 5,126 screened

none of the four intervals crosses zero

A preregistered review with a meta-analysis. Only studies measuring all three quantities were included: the human alone, the model alone, and the pair.

The headline figure for the whole review is minus 0.23: on average, the pair is worse than the stronger of the two sides. From this figure, both halves of the review read as parts, not as the final verdict.

Where the model is stronger than the human, both figures are measured on the exact same studies: the pair is worse than the model by 0.54, while also beating the human alone by 0.74. Both together are the rule: the pair outperforms the weaker side and doesn't catch up with the stronger one.

Across the whole review, the pair beats the human alone by 0.64 — but that's a different base and different studies, and this number doesn't go next to minus 0.54.

Vaccaro, Almaatouq, Malone, Nature Human Behaviour 2024

Example 4: surgeons assess risk — on their own and after consulting the model

UNDERESTIMATING THE RISK

the patient is operated on as a routine case

no ICU bed ready, no conversation with the family

OVERESTIMATING THE RISK

the operation gets delayed or cancelled

it's invisible in the thirty-day count: no surgery means nothing to compare

AREA UNDER THE CURVE, SURGEON ALONE → WITH POTTER

mortality 0.841 → 0.870 · septic shock 0.820 → 0.712

pneumonia 0.753 → 0.803: on three of the four outcomes, the pair is more accurate than the surgeon alone. Scale: 0.5 is a coin flip, 1.0 is no misses

Surgeon alone

written case description ➡ risk assessment ➡ five outcomes

Surgeon after the model

case description ➡ **POTTER: 4–11 answers** ➡ risk assessment ➡ five outcomes

Randomized into two groups of 15; no one examined the patient in person

30 surgeons assessed risk from 150 written case descriptions; outcomes were tracked forward for 30 days.

On septic shock, the pair comes in below both the surgeon and the model: the model alone scored 0.816. **The second classical mode — risk scales,** a different sample: 0.874–0.898 against the model's 0.916.

Same performer, now worse. Rung — looks second. Where the model's lead was largest, the pair added nothing at all: **0.834 against 0.833.**

Bertsimas et al., Annals of Surgery 2018 · *the model against surgeons' judgment, Journal of Trauma 2023* · *Everett et al., npj Digital Medicine 2026*

The answer, assembled: four cases, four effects, one conclusion

case	effect	conclusion
Surgeons and POTTER. 30 surgeons, 150 cases, randomized	on three of four outcomes the pair is more accurate than the surgeon; on septic shock **0.712** against 0.820 and 0.816	lost exactly where the human was stronger
Physicians with a language model. 50 physicians, randomized	reasoning score **76 versus 74**: a 2-point difference, interval −4…8	indistinguishable from a reference manual
A review with a meta-analysis. 74 studies, 106 experiments	where the model is stronger — the pair is worse than the model by **0.54**, better than the human by 0.74; across the whole review, −0.23	the scoring runs against the stronger side
The eye-tracking study. 10 readers, against themselves	where the suggestion stayed silent (14 cases) — missed cancers **29** → **61** out of a hundred	the sign is set by whether the suggestion was right

In none of the four did the human's decision actually reach a patient. The outcomes serve as the reference standard for scoring accuracy, not as a consequence of the suggestion.

surgeons: Journal of Trauma 2023 · *physicians with a language model: JAMA Network Open 2024* · *the review: Nature Human Behaviour 2024* · *eye-tracking: Radiology 2026*

Example 5: an early-warning model for sepsis

THE COST OF A MISS

hours until the ICU

not measured in money

THE COST OF A FALSE ALERT

after a hundred of these — dismissing without reading

time taken from another patient

CONFIRMED OUT OF ALERTS RAISED: SCALE → MODEL

8 → 12 out of a hundred

scale: 45,776 patients, sepsis in 3.3 out of a hundred · model: 38,455 admissions, 6.6 out of a hundred

CLAIMED → MEASURED BY AN INDEPENDENT VALIDATION

0.76–0.83 → 0.63

area under the curve: 0.5 is a coin flip, 1.0 is no misses. Caught 33 cases out of a hundred — missed 1,709 out of 2,552

Scale by hand	points from a table	➡	the sum	➡	the threshold	➡ the assigned action
Model in the record	vitals and notes, continuously	➡	**risk score**	➡	alert with orders	➡ the shift reviews it

The classical mode isn't zero, it's a scale tallied by hand

Wong, Otles et al., JAMA Internal Medicine 2021, with an erratum in the same journal · a comparison of scoring scales, JAMIA Open 2021 · Ostermayer et al., JAMIA Open 2024 · claims for the second version and a prospective validation

Example 5, the cost: eighteen alerts per hundred admissions

TO FIND ONE CASE, REVIEW

eight alerts

an earlier figure of fifteen was retracted by an erratum in the same journal

A SECOND VALIDATION, DIFFERENT AUTHORS AND HOSPITALS

lead time — zero minutes

median; 145,885 visits, 2023

SHARE OF PATIENTS WITH AN ALERT DURING THE EPIDEMIC

9 → 21 out of a hundred

a different base — 24 hospitals: alert stream 953 → 1,363 while census fell 10,159 → 6,634

Before	scheduled rounds	a score from the table	call the physician or not
After	the record computes continuously	**the model's alert**	review and dismissal

Reviewing an alert — a new step for the same shift

An alert arrives for 18 out of 100 admissions; 12 out of 100 raised alerts are confirmed: counted from different things — don't mix them up.

The scale was confirmed 8 out of 100, the model 12: it never left the classical band. The populations differ: this compares order of magnitude, not superiority.

Verbatim: "adds little to what physicians are already doing."

What shifted: the load grew at the same spot; **a step appeared** — with the scale there was nothing to dismiss, since it computed its own threshold. **The input distribution shifted; the threshold stayed put. Alert stream, the fourth input distortion.**

Wong, Otles et al., JAMA Internal Medicine 2021, with an erratum in the same journal · *Ostermayer et al., JAMIA Open 2024* · *alert volume during the epidemic*

Example 6: Moscow, Russia — a model suggests a final diagnosis from the medical record

PHYSICIAN ALONE → MODEL

91.7 → 87.7 correct out of a hundred

3,332 cases: the model is four points WORSE than the physician, and it's deployed as insurance against an overlooked detail

THE CODE SET

256 → 571 → 363

571 covered 95 out of a hundred visits; it was cut back by removing codes a general practitioner never assigns

THE MODEL CAN STAY SILENT

5 million requests → 3 million answers

above the uncertainty threshold, there's no answer at all

Physician alone	complaints and exam	physician builds the list	physician chooses	code goes into the record
Moscow, with the model	the whole record	**model suggests a code**	physician decides whether to trust it	code goes into the record

The suggestion arrives BEFORE the physician's decision and sits above it on screen

What kind of case this is. Moscow, a general practice visit. The model reads the whole medical record and, before the physician's decision, suggests a final diagnosis as a code. *The product is AIDA.*

How it's built: input — the whole record: complaints, history, exam, labs, reports, the code history. Output — a single code with an uncertainty score. A Longformer up to 8,192 tokens, 1.5 million records. *Text processing.*

Umerenkov, Nesterov, Shaposhnikov et al., IJCAI-25, AI for Social Good track, pp. 9880–9886

Example 6, the cost: where the physician and the model disagree

AGREE — 2,635 CASES

99 correct out of a hundred

the easy part of the set: nothing to decide

DISAGREE — 697 CASES

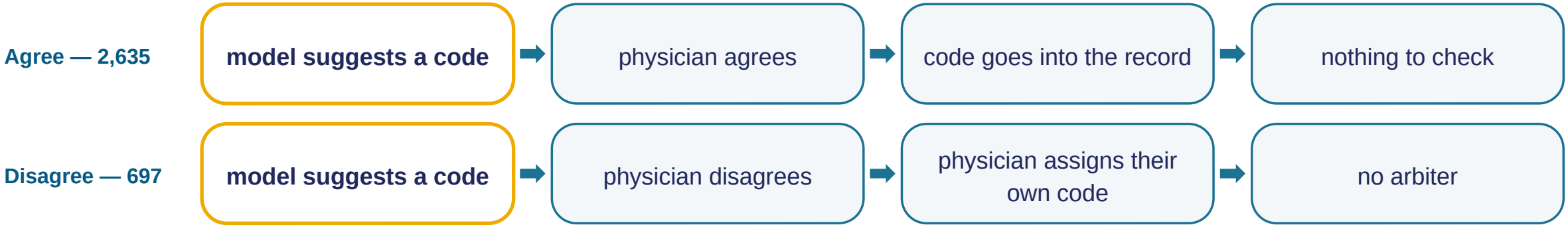
physician 64 → model 45 out of a hundred

in one case out of five, neither is reliable: the physician gets it right less than two-thirds of the time, the model less than half

THREE EXPERTS DIDN'T FULLY AGREE — 965 CASES

physician 78.2 → model 72

both collapse together, by twenty points



The same physician resolves the disagreement — there's no third reader at this stage

Where they agree, both are right and there's nothing to decide. All the cost sits in the fifth of cases where they disagree.

The reference standard itself is unstable: on 168 of 3,500 cases, three experienced physicians didn't agree on anything at all.

Umerenkov, Nesterov, Shaposhnikov et al., IJCAI-25, Table 5, p. 9882

Section conclusion: one tool, opposite outcomes

What works

- a second look at the whole stream
- a reminder of a rare diagnosis
- the pair stronger than both sides — under one design of the procedure
- arbitration only on disagreements: the whole stream doesn't go to it

What doesn't work

- preserving the independence of two opinions
- holding onto the advantage where the human is stronger
- noticing that the input distribution has shifted
- assigning a rung of involvement: no one writes it down

Six case studies at this stage produced opposite outcomes with the same tool. The Swedish trial: one human reading on top of the model beats double reading. The surgeons: the same pair falls below both. What differs is the design of the procedure — whether the independence of the two opinions is preserved, and the order in which the suggestion is presented.

What holds this up: a threshold set against the cost of both errors, and arbitration on disagreements.

The rung of involvement is written down nowhere. Left unassigned, it slides down to signing without looking.

Of the six input distortions, this section met four: anchoring on an incorrect readout · narrowing · alert stream · disagreement as a standing step.

4. Keeping the medical record

Observation looked for deterioration in numbers. Here, the decision moves into text, and it will be read by someone who never saw the patient. **A gap in the record and the absence of an event look the same to them.**

• **the visit note** — read by the patient themselves and the next physician

• **the discharge summary** — read by the local physician, the next hospital, the insurer

In order, and the case study first:

Carrying forward a previous note: the heparin case and the four writing modes

Example 1: automated visit-note generation

Example 2: the discharge summary

Example 3: voice dictation of a radiology report



The record: the data, the question, the decision

TYPED BY HAND

18% of the note's characters

23,630 notes, 460 authors at one center

CARRIED FORWARD · AUTO-POPULATED BY THE SYSTEM — SHARE OF CHARACTERS

46% · 36%

same sample; among residents, carrying forward runs at 51.4%

The data for this stage is everything gathered earlier, put into text. The stage adds no measurement of its own.

Five ways a note gets produced: typed by hand with the patient present · typed after the visit · dictated · auto-populated from the record's own fields · carried forward from a previous note.

The decision this stage makes is what goes into the document that treatment will later be ordered from. It comes after observation: the record gets read later, and a different person reads it, carrying their own question.

Wang et al., JAMA Internal Medicine 2017

What happens at this stage: the visit note and the discharge summary

The visit note	Hold the structure in mind during the conversation	Type up the note after the visit	Proofread and sign
The discharge summary	Assemble the summary from the progress notes	Hand the document to the next physician	Produce it if the case is reviewed

Six actions of the stage, three per document form

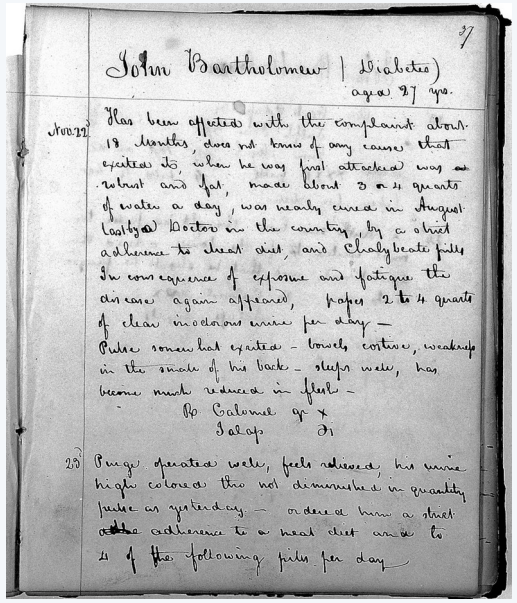
The visit note. They listen to the conversation; the text holds the complaints, exam findings, diagnosis, and treatment plan. It's read by the patient themselves and the next physician.

The discharge summary. There's nothing to listen to — it's assembled from the progress notes covering the whole hospitalization. It's read by the local physician, the next hospital, and the insurer.

Why keep medical documentation at all

A decision about a person outlives the person who made it. The physician will go on leave, change jobs, forget; the patient will turn up a year later at a different institution. The only thing that crosses between them is the text.

A case book, 1819. Name, age, diagnosis, how long the complaint has lasted, what was prescribed, an entry for the next day.



Two hundred years later, these are the same fields — only now their composition and the order they're kept in are set by a government order, not by a physician's habit.

The case book of the Philadelphia Almshouse infirmary, 1819–1820, Wellcome Collection L0016766 · Russian Ministry of Health order No. 530n of 05.08.2022

Carrying forward a previous note: the heparin case and the four writing modes

A patient on chemotherapy: a note on heparin prophylaxis

- **Day 1** — A note on heparin prophylaxis goes into the progress note. **No heparin was actually ordered.**
- **Days 2–5** — On transfer to another department, the note is carried forward from note to note. Everyone who reads it assumes the prophylaxis is being given.
- **After discharge** — **Pulmonary embolism, readmission.**

CARRIED FORWARD VERBATIM — SHARE OF THE CHART'S WORDS

50.1%

104 million notes on 2 million patients

CONTRIBUTION OF CARRY-FORWARD ERRORS

2.6% of 190 errors

190 across 212,165 visits

Counted	The physician writes it themselves — minutes per note	Carrying forward a previous note — share of the chart's words
Named, no figure	A live scribe	A resident's draft

Four writing modes. Who puts the text into the document

Tsou et al., Applied Clinical Informatics, 2017 · Steinkamp et al., JAMA Network Open 2022

What the model took over at the record-keeping stage, and what it still can't do

OF THE SIX ACTIONS, THE MODEL TOOK OVER

two

typing and assembling the document; it removed one — holding the structure in mind

Works: typing up the note and assembling the discharge document. Holding the structure in mind during the live conversation has been taken off the person — the one place in this lecture where a step was actually removed for them.

Doesn't work, in three places: removing the obligation to proofread · cutting active editing time · telling the plausible apart from the actually recorded.

Whoever decides from the record never saw the patient: a gap and the absence of an event look the same to them.

- **record less than happened** — something already checked gets ordered again
- **record what never happened** — treatment gets ordered on a false basis

Example 1: automatic recording of the visit

BASELINE: THE PHYSICIAN WRITES IT HIMSELF

4 min 22 s per note

control group; its own drift — 18 seconds

FIRST TOOL AGAINST THE CONTROL GROUP

−9.5% of note time

ACTIVE EDITING TIME

did not change for either tool

CLAIMED BY THE VENDOR, NO METHODOLOGY GIVEN

15,791 hours on 63,000 visits

a quarter hour per visit does not fit inside four minutes

Without the model

Conversation → Structure held in the head → The physician types it up himself → Signature

With the model

Audio of the visit → **The model produces a draft** → Reviewing someone else's text → Signature

The same visit, two orders of work: without the model on top, with it below

The first tool to reach this stage was one that writes the note itself. *Text generation: audio of the visit goes in, a draft note comes out.*

How it was measured. 238 physicians, three equal groups: two tools and ordinary practice. **The second tool showed no change** — the effect belongs to the implementation, not the class.

A three-arm randomized trial, medRxiv preprint 2025

Example 1, Russia: a voice assistant fills out the visit note

MOSCOW PRIMARY CARE PHYSICIANS WITH THE ASSISTANT

more than 3,000

all adult city polyclinics, since September 2026

NOTE QUALITY, PER EXPERT RATING

4.8 out of 5

the same as a note filled out by a person

PHYSICIAN'S ATTENTION TO THE PATIENT

more than doubled

among neurologists; the starting figure is not stated

Without the model

Conversation → The physician types it up → Signature

With the model

Conversation → **The assistant sorts it into fields** → The physician reviews and corrects → Signature

The same visit, two orders of work — a domestic deployment

Speech recognition: the visit conversation goes in, the note's fields come out.

How it was measured. Expert rating of finished notes. No blinding and no control group.

Trained on de-identified recordings of real visits.

Moscow City Health Department, 09.09.2026 · *Moscow City Health Department, 25.06.2026*

Example 1, cost: typing shrank, review appeared

GAVE

note finished earlier, less fatigue

REQUIRED

reviewing every draft, and a staffed department

DEPENDS ON THE DAY OF MEASUREMENT

7 → 16 out of 100

observational; day of rollout → day 150

NET

typing shrank, editing stayed the same

Before	typing	➡	signature			
Now	conversation	➡	draft	➡	review	➡	signature
On top, by another hand	spot-checking of notes					

The physician's two tracks — and a third line: an operation someone else does

The physician used to hold the note's structure in his head with the patient there, type it up after the visit, and sign his own work. Now he reviews someone else's.

~~**The control department's staffing — from an unpublished clinic report: this number has no address of its own.**~~

A quality-control report on notes from one clinic · [a three-arm randomized trial, medRxiv preprint 2025]

Example 2: the discharge summary

AT LEAST ONE ERROR IN THE DOCUMENT

9 of 11 machine-written summaries

of 19 fabrications, 10 were major

COMPLETENESS: HUMAN → MODEL

4.13 → 3.72

a five-point reviewer scale

UNIQUE ERRORS PER DOCUMENT

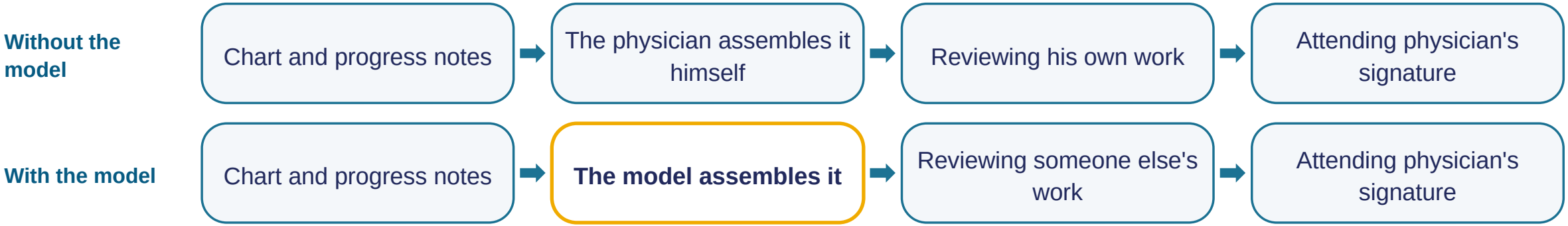
1.82 → 2.91

human → model

POTENTIAL HARM OF A SINGLE ERROR

1.34 → 1.35

out of seven; not distinguishable



Assembling the discharge document, two orders of work

Text generation: the medical record goes in, a draft of the narrative section comes out.

How it was measured. 100 hospitalizations, 22 reviewers: each document was read blind by two of them.

Shorter and more coherent, but less complete. The study names neither the model nor its training data.

The cost of error shifted: the same person writes it, but the reader pays.

A blinded comparison of 100 hospitalizations, JAMA Internal Medicine 2025

Example 2, cost: a different kind of review

MACHINE-GENERATED CONTENT WAS USED

in 219 of 384 discharges

there were 1,274 summaries

PER HUNDRED SUMMARIES

25 omissions · 20 inaccuracies · 2 fabrications

OF A HUNDRED UNEDITED SUMMARIES

88 with no potential for harm

1 moderate · 0 severe

THE TRIO OF HARM FIGURES THAT REVIEWS CITE

is not computed in any of the three studies

At night

Chart and progress notes → The model assembles a summary

During the day

The physician reads it if he chooses to → Looks for what wasn't there → Signature

Where review sits and what it looks for

The discharge summary was written and signed by the attending physician; at teaching hospitals a resident prepared the draft — a mode with no number attached.

Correction, the sixth input distortion, works in our favor. Review changed its target: the eye is tuned to catch a typo, but what it has to look for is a statement that was not in the medical record. **Nobody teaches this kind of reading.**

A prospective pilot, JAMA Network Open 2026 · *Cureus 2026*

Example 3: voice dictation of the radiology report

BASELINE: KEYBOARD ENTRY

10 min 15 s per study

one CT or MRI report

SPEECH RECOGNITION, SAME MEASUREMENT

8 min 2 s

−2 min 13 s, 21.6% of the baseline

SCALE AS OF JANUARY 2023

210,000 reports

the vocabulary trained on 2.5 million reports

Without the model	Image on screen	➡ Typing on the keyboard	➡ Editing and signature	➡ Report in the chart
With the model	Speech into the headset	➡ **The model recognizes it**	➡ Editing and signature	➡ Report in the chart

Reporting a single study, two orders of work

Speech recognition: the physician's speech goes in, the report's text comes out.

For forty years this technology lost to the keyboard: in 1981, voice input took four times as long and produced 12 errors against zero.

How it was measured. Radiologists at seven polyclinics reported studies both ways; time was measured by stopwatch.

Kudryavtsev et al., Digital Diagnostics 2023 · *Moscow City Health Department, 07.01.2023*

Example 3, cost: word endings, other people's speech, and a number with no base

REPORT AN INCREASE IN THEIR OWN EFFICIENCY

62.8% of those surveyed

2022 survey of radiologists; self-report, not a measurement of time

SECOND DEPLOYMENT: TWO ENDOSCOPY CENTERS

−27% of report time

180–200 studies a day per center; HOW MANY MINUTES IT USED TO TAKE — not published

What was said	word endings	other people's speech in the room
What it was recorded with	equipment noise	a cheap microphone

Two independent sources of result spoilage — and neither is in the model

The physicians most willing to take up the tool are those aged 30–40.

62.8 percent is the share of physicians who THINK SO, not a measured time.

The second deployment claims −27% of report time. From how many minutes? Say it out loud — and we'll check it against what was published.

Kudryavtsev, Bardasova, Khoruzhaya, Digital Diagnostics 2023;4(2) · NIIOZMM (Moscow City Health Department's Research Institute for Healthcare Organization and Medical Management), "Moskovskaya Meditsina" ("Moscow Medicine"), endoscopy centers, 2022

Dictation that suggests a diagnosis: authorization and measurements

FOREIGN SYSTEMS CHECKED

16

authorizations to suggest from speech — zero; independent accuracy measurements — zero

DOMESTIC NAMES WITH VOICE INPUT CHECKED

13

not one "speech → suggestion" system among them

MOSCOW TOP-3: PHYSICIAN AGREEMENT WITH THE SUGGESTION

46% · 68%

with one diagnosis offered and with a choice of three; it has no voice input

The medical-device boundary is drawn at the signature. Suggesting a **code** for the physician to check is documentation. Suggesting a **diagnosis** for the physician to check **is already a medical device**, even when the check is mandatory.

Glass Health and Corti do this now — they build out a presumptive diagnosis as live speech unfolds. *Neither has an accuracy measurement.*

46% is physician agreement with the suggestion, not its accuracy.

UK framework for ambient voice technology-enabled products, 29.07.2026 · *TOP-3*

Section conclusion: what works, what doesn't, and where the load moved

THE ENTIRE MEASURED SAVINGS OF THE STAGE

9.5% of note time

against the control group, 4 min 22 s; absent for the second tool

ACTIVE EDITING TIME

did not shift for either tool

DOCUMENT COMPLETENESS: HUMAN → MODEL

4.13 → 3.72

a blinded comparison of 100 hospitalizations

DOMESTIC TIMING STUDY OF THE REPORT

10 min 15 s → 8 min 2 s

keyboard versus speech

What works. The draft is ready by the end of the visit; the structure no longer has to be held in your head while a living person is in front of you — the one step that was actually removed.

What doesn't. The model does not write more completely than a human. The claimed 15,791 hours have nothing to confirm them.

The load moved from typing to reviewing someone else's text; spot-checking of notes appeared, with its own staff. Next in the cycle is what holds that review in place: **the signature.**

Nobody teaches this other kind of review anywhere.

Of the six input distortions, the section met one — correction: the physician corrects the machine's work.

5. Professional qualification: maintenance and erosion

What happened to the ones who decide. The lecture has been measuring the work. What happened to the person inside that work is told by three measurements — and each has its own weakness.

3 breakdowns · 2 classical modes with numbers · a review of authorization across three countries

Example 1: developers on an unfamiliar library — the second half of the pair, measured by lottery, and taken outside the clinic.

Example 2: the rate fell without the assistant — the only measurement of erosion on real clinical work.

Example 3: ArtInCol, Russia — an assistant on the same procedure — the same quantity, a stronger trial, the drain not measured.

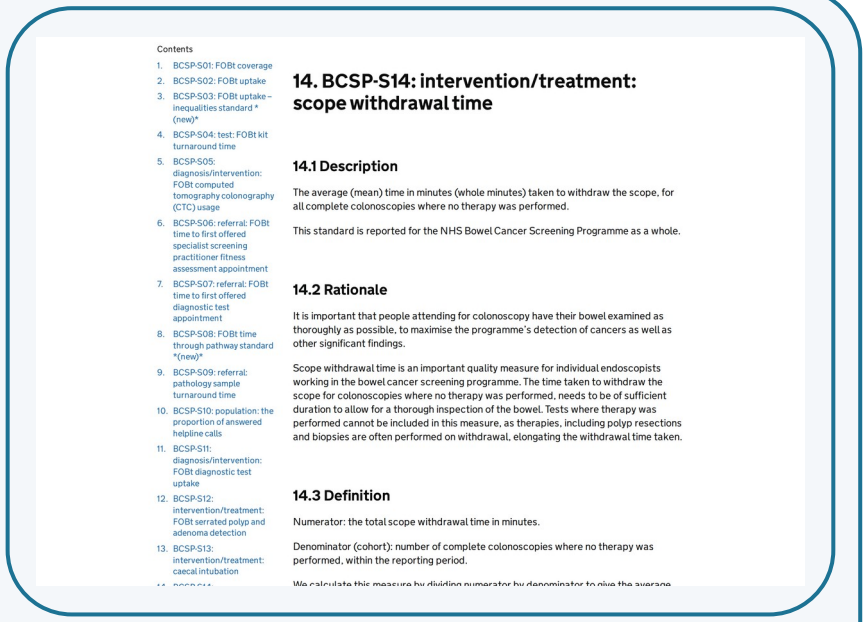


Qualification: what is measured, and how

Here, what gets measured is the performer himself: is he in form. Skill is spent in silence.

An adenoma is a precancerous thickening of the mucosa: removed today, it will not become cancer years from now.

Data is gathered four ways, and all four run on the ordinary flow of work: an audit of the decisions made on the job — the rate of procedures with a finding · timing by stopwatch · a count of procedures for the year · periodic re-certification. **Case review comes after the outcome and is not part of the count.**



Contents

1. BCSP-S01: FOBt coverage
2. BCSP-S02: FOBt uptake
3. BCSP-S03: FOBt uptake – inequalities standard * (new)*
4. BCSP-S04: test: FOBt kit turnaround time
5. BCSP-S05: diagnosis/intervention: FOBt computed tomography colonography (CTC) usage
6. BCSP-S06: referral: FOBt time to first offered specialist screening practitioner fitness assessment appointment
7. BCSP-S07: referral: FOBt time to first offered diagnostic test appointment
8. BCSP-S08: FOBt time through pathway standard *(new)*
9. BCSP-S09: referral: pathology sample turnaround time
10. BCSP-S10: population: the proportion of answered helpline calls
11. BCSP-S11: diagnosis/intervention: FOBt diagnostic test uptake
12. BCSP-S12: intervention/treatment: FOBt serrated polyp and adenoma detection
13. BCSP-S13: intervention/treatment: caecal intubation

14. BCSP-S14: intervention/treatment: scope withdrawal time

14.1 Description

The average (mean) time in minutes (whole minutes) taken to withdraw the scope, for all complete colonoscopies where no therapy was performed.

This standard is reported for the NHS Bowel Cancer Screening Programme as a whole.

14.2 Rationale

It is important that people attending for colonoscopy have their bowel examined as thoroughly as possible, to maximise the programme's detection of cancers as well as other significant findings.

Scope withdrawal time is an important quality measure for individual endoscopists working in the bowel cancer screening programme. The time taken to withdraw the scope for colonoscopies where no therapy was performed, needs to be of sufficient duration to allow for a thorough inspection of the bowel. Tests where therapy was performed cannot be included in this measure, as therapies, including polyp resections and biopsies are often performed on withdrawal, elongating the withdrawal time taken.

14.3 Definition

Numerator: the total scope withdrawal time in minutes.

Denominator (cohort): number of complete colonoscopies where no therapy was performed, within the reporting period.

We calculate this measure by dividing numerator by denominator to give the average

The finding rate cannot be raised by trying harder: an adenoma lands in the denominator even when the physician did not look behind the fold. Withdrawal time is a discipline — easier to keep up.

The decision: authorize, narrow the authorization, suspend it, or not renew it.

Why this stage stands here: erosion accumulates across all stages at once — it can only be gathered once all of them have passed.

Of the four methods, one measures the result; the rest measure behavior and volume. Which one — and why is it the easiest to substitute with the other two?

The image shows BCSP-S14 standard "Scope withdrawal time" from the national bowel cancer screening programme, GOV.UK · colonoscopy quality indicators, Gastrointestinal Endoscopy 2015

The physician came back after a year: what keeps him in form, and where the model landed

The physician came back after a year with no procedures — an administrative post, illness, parental leave. Same diploma, authorization never revoked: there was nothing to revoke it for.

The first procedure goes differently: the scope moves slower, the hand takes longer to find the angle, he passes a fold by. No complications, the report is written, **and he doesn't know it.** A bad examination looks exactly like a good examination in which nothing was found.

That is why three pillars keep him in form, and none of them asks the person himself: volume of practice for the year, authorization from the organization, periodic re-certification.

Russia: periodic accreditation once every five years — 144 hours of study plus a portfolio. **The specialist writes the work report himself. There is no numeric threshold on the result — neither a finding rate nor a withdrawal time — in the requirements.**

The model took over one action out of eight — the withdrawal examination. **One was added: whether to trust the box.**

Professional society guidelines, Gastrointestinal Endoscopy 2015 · Order No. 709n of the Russian Ministry of Health, 28.10.2022, "On approval of the Regulation on the Accreditation of Specialists"

Example 1: developers on an unfamiliar library

WROTE BY HAND: CORRECT ON THE TEST

67 out of 100

the gap is wider on "find the bug" tasks

WITH THE ASSISTANT: THE SAME TEST

50 out of 100

effect size 0.738, significance 0.01; no gain in time

ASKED ONLY ABOUT CONCEPTS

above 65 out of 100

7 people; those who handed the task over whole — four of them — below 40 out of 100



What was compared with what

Skill is spent in silence — and so is skill that hasn't formed yet. What was done: 52 participants — junior engineers unfamiliar with an asynchronous-programming library — were split by lottery: half with the assistant, half by hand. Everyone was tested without the assistant.

The subject being measured here is programming, not medicine. The second half of the pair — the result without the assistant — is measured by lottery: neither of the two physician breakdowns that follow will measure it this way.

What is measured here is a skill that never formed: nobody knew the library, so there is nowhere for erosion to come from; the glossary has no form for a case like this.

The same drain happens with no model at all. In UK cytology, the volume of reading dropped 80–85 out of 100 after the switch to human papillomavirus testing, and the number of centers fell to 8 out of 45. **It is the handed-off volume of work that drains skill, not the model.**

Shen and Tamkin, arXiv:2601.20245, 2026 · *Heudel et al., ESMO Real World Data and Digital Oncology 2026*

The withdrawal examination: the cost of a miss, and two orders of work without the model

MISSING AN ADENOMA

stays in the person until the next examination

SEEING TOO MUCH

removal, bleeding, a repeat visit

PROCEDURES WITH A FINDING: THE SCOPE IS WITHDRAWN MORE SLOWLY

271 → 366 per 1,000

1,027 patients, by lottery; nine minutes instead of six

PROCEDURES WITH A FINDING: A NURSE WATCHES THE SCREEN TOO

393 → 457 per 1,000

1,676 patients, a pooled analysis of three trials

Ordinary order	insert the scope	examine during withdrawal	check behind the fold
Two points of intervention	the timer: not six minutes but nine	a nurse watches the screen too	

Where intervention happens without the model

For physicians the same check begins with the cost of a miss. A bad examination looks like a good one in which nothing was found — the physician has no signal of his own.

Both gains are of the same order as the assistant's, or larger, and neither is free: three minutes of the physician's time per procedure, and a second person in the room.

The baselines differ: 271 per 1,000 for the first order of work, 393 for the second — different hospitals and patients. The gains do not add up.

Zhao et al., Clinical Gastroenterology and Hepatology 2022 · *Lei et al., International Journal of Colorectal Disease 2024*

Does the physician's measured work result count toward his authorization: the UK, the US, Russia

UK: THE THRESHOLD COUNTS TOWARD AUTHORIZATION

15 procedures with a finding out of 100, and above

the minimum standard; 280 colonoscopies before certification, 100 a year after

US: THE SAME QUANTITY, FORBIDDEN TO BE USED AS AUTHORIZATION

target level of 35 out of 100

the profession's own registry has gathered more than 10 million colonoscopies

RUSSIA: THE SPECIALIST WRITES THE WORK REPORT HIMSELF

no threshold on the result · 144 hours of study over 5 years

neither a finding rate nor a withdrawal time

Both orders of work from the previous slide are measured by the rate of procedures with a finding. That same rate also underlies authorization.

The quantity is the same across three countries; the consequence diverges. In the UK, it is used to authorize work in screening. In the US, the very document that set the target level calls applying its own indicators as a condition of authorization a misuse. In Russia a work report is required, but the specialist writes it himself: there is no threshold counted on his behalf.

We looked and found this in none of the three: a renewal of authorization that would factor in a physician's measured result **after a period of working with highlighting**.

Siau et al., Frontline Gastroenterology 2023 · ASGE/ACG, Gastrointestinal Endoscopy 2024;100(3) · Order No. 709n of the Russian Ministry of Health

Example 2: the rate fell without the assistant

BEFORE THE ASSISTANT

284 procedures with a finding per 1,000

226 of 795

AFTER THREE MONTHS

224 per 1,000

145 of 648

RESULTED IN

minus 60 findings per 1,000

19 physicians, average 28 years of experience

With the assistant	the stream of frames	**a box on the region**	the physician decides for himself
What was measured	the same physicians	procedures WITHOUT the assistant	

Where the assistant sits, and where the physician was measured without it

What was done. At four centers in Poland, endoscopists were given an assistant that boxes a finding. Three months later, the same physicians' procedures were compared **without** it.

This is a cognitive distortion — skill decay, the third of the six presented at the opening of the evening. In a person it shows up like this: the same physicians, without the assistant, find less.

Gave: they find more while it's on. **Required:** the skill of searching on your own. **Net:** the performer is the same — the place, the volume, and the hand are unchanged — but got worse.

The publication does not name its own system.

Budzyń et al., Lancet Gastroenterology & Hepatology 2025 · *Bastani et al., PNAS 2025*

Example 3: ArtInCol, Russia — an assistant on the same procedure

WITHOUT THE ASSISTANT

413 procedures with a finding per 1,000

547 patients

WITH THE RUSSIAN ASSISTANT

472 procedures per 1,000

581 patients

WHAT THE TRIAL LACKS

a measurement after the assistant is removed

Mechanics

video stream from the scope → a computer by the cart → **a box on the monitor**

Trained on

the archive of the National Medical Research Center of Coloproctology → annotation by its own endoscopists

Where the system sits in the workflow

What was done. At four institutions in Russia — Moscow and Ufa — 1,128 patients were split by lottery: colonoscopy as usual, and with a Russian assistant that boxes a region.

Two glossary forms. The performer changed: part of the search is done by the authorized device. The load grew at the same spot: 1.23 false alarms per study where there used to be zero — that many times the physician looks at an empty spot.

The Russian deployment is the largest in the world by claimed numbers, yet there is no measurement of the consequences: the physician was not measured here after the assistant was removed.

Achkasov S.I., Shelygin Yu.A. et al. Efficacy of the ArtInCol artificial-intelligence system in the diagnosis of neoplasms during colonoscopy: results of a multicenter randomized study. Koloproktologiya, 2025; 24(3):12–21

Two modes of one class: where the person stays, and where he is removed

HIGHLIGHTING: THE PLACE STAYED WITH THE PERSON, THE DRAIN IS MEASURED

284 → 224 findings per 1,000

the same 19 physicians, procedures without the assistant

SECOND READER: THE PERSON WAS REMOVED FROM THE SPOT

human reading down by 44.3 percent

the Swedish program, 80,033 screened

THE DRAIN AT THE REMOVED SPOT

is not measured and cannot be measured

there's nobody to measure

Video highlighting	the physician performs the procedure	**a box on the region**	watches and decides for himself
Second reader of the scan	the scan is ready	**the model's reading**	there is no second person

Two modes, and what is left to the person

Both clinical breakdowns in this section are taken on one mode — highlighting. There are two modes, and the difference isn't in accuracy.

Video highlighting: the physician performs the procedure, a box appears on the region, the spot stays with him — and the drain at that spot is measured. **Second reader of the scan:** the model takes the second person's spot, and the person is removed from that spot entirely.

Skill is drained where the person formally stayed. Where he was removed, the drain cannot be measured. This is a property of the decision itself, and no future measurement will close that gap.

Budzyń et al., Lancet Gastroenterology & Hepatology 2025 · *MASAI safety analysis, Lancet Oncology 2023*

Stage summary: what works, what doesn't, and what this means for you

FOR PHYSICIANS

284 → 224 findings per 1,000 procedures

observational, once

IN EDUCATION, BY LOTTERY

exam with no model
17% below the control

about 1,000 high-school students

A COUNTERWEIGHT WITH NO NUMBER OF ITS OWN

for 69 novices, a week later, the drain was not confirmed

a one-week horizon; the source does not state the size of that "not confirmed"

What works: they find more while it's on; measured by lottery and replicated.

What doesn't: telling the physician about a miss inside the procedure · measuring trust over months.

A step appeared: working out disagreements between the physician and the model. Routine re-checking of the flow has been replaced by targeted arbitration: the defense changed completely.

No procedure in any country measures a physician without the assistant.

About you: an unrestricted assistant creates an illusion of learning: higher during practice, lower on the test. It's cured by the design of access, not the amount of practice.

Of the six input distortions, the section met one — skill decay: minus 60 findings per 1,000.

Budzyń et al., Lancet Gastroenterology & Hepatology 2025

6. Wrap-up

The lecture examined cases one at a time. Here they add up into a tally: each summary answers its own question.

1. **The nine forms in full: what happened to the work**

2. **The count of misattributions: in thirteen of twenty-one cases, the win does not belong to the model**

3. **The map of tasks and data types: four cells out of twelve, and what works in them**

4. **What was paid: attention, skill, the signature, arbitration**

There is one new number here — the count of misattributions. The section adds not a single new case: everything else has already been presented, each with its own base, and it is combined here so that the sum reveals what no single case shows on its own.

There is no answer here to the question the evening began with: the closing section opens with one line about what the evening did not measure, and answers it on the next slide.

The nine forms in full: what happened to the work

	form	tally
1	The load moved to another step	note time **−9.5%** vs. control, editing unchanged
2	The step disappeared entirely	**no case encountered**
3	A step appeared	arbitration **3.9** → **9.4** and **11.3** → **13.8** out of 100 reads
4	The load grew at the same spot	**18** alerts per 100 hospitalizations · **1.23** wasted per study
4a	The load fell at the same spot	reading down **44.3%**
5	The performer changed	turnout **22** → **100** out of 100 · findings **413** → **472** per 1,000
6	*The order of work changed*	**no tally**
7	The cost of error shifted	on dark skin, worse **by 9–15** out of 100 · more chronic conditions **by 26** out of 100
8	The performer is the same, but got worse	findings **284** → **224** per 1,000
8a	*The performer is the same, but got better*	*no measurement*

Seven of the nine forms carry a tally; both dashes are named as dashes. Which of the nine, in your view, came up most often this evening — and which never came up at all?

the tally is compiled from the lecture's breakdowns

The count of misattributions: in thirteen of twenty-one cases, the win does not belong to the model

NO MODEL IN THE MECHANISM AT ALL

2 of 21

remove the model and the mechanism still runs: an "if-then" decision tree in triage; skill is drained by the handed-off volume of work

THE WIN DOES NOT BELONG TO THE MODEL

7 of 21

belongs to the threshold, the training-set composition, the target set for training, the order of presentation, or the implementation

THE NUMBER GREW IN THE RETELLING

4 of 21

56.7% instead of zero · "no effect" collapsed into one number · a share of opinions instead of time · scale instead of measurement

Twenty-one breakdowns all evening. In thirteen, the win or the blame belongs not to the trained model.

Seven breakdowns out of twenty-one are domestic practice: mammograms in screening, Moscow twice in diagnostics, the chart in diagnosis, the visit note and voice dictation in documentation, ArtInCol in qualification.

Three strong lines of this through-line never made it to the screen: their sections were cut from the lecture. The strongest — a peer-reviewed review that got six of its own seven primary sources wrong.

The practice to take from this: for a claimed effect, ask **what it's counted against**; for a retold one, **open the primary source**.

the tally is compiled from REESTR-PRIPISOK-DEKI.md

The map of tasks and data types: four cells out of twelve, and what works in them

Four rows of tasks against three data types — twelve cells. Four are named; nobody went into the remaining eight across five sections.

task, and where it came up	what works	what doesn't
classification — image: screening, diagnostics	a standardized scan, fitness-for-analysis settled before the model	the subgroup underrepresented in the training set: it pays
recognition — image: screening, diagnostics, diagnosis	a flow filter: who gets read twice	grounds to drop a reading: savings from what goes unread
retrieval and generation — text: screening, diagnosis, documentation	a draft ready by the end of the visit	more complete than a human: the same person writes it, but the reader pays
forecasting — structured data: diagnostics, diagnosis	a signal alongside the scale	the decision: the target variable sets the aim, and a document sets the threshold

Three more spots lie outside the window. Speech at the visit: typing by voice works; word endings and other people's speech do not. Video: highlighting during the procedure, with the spot staying with the person. Instrument signal: there is no trained model at all, yet there is bias.

The right column, in not one row, is about the quality of the model — it is about the fitness of the input, the composition of the data, the target set for training, and the assigned threshold. **Qualification occupies nothing in the window:** its assistant works on video.

What was paid: attention, skill, signature, arbitration

The four payments from the start — by name, with a measurement under each.

payment	what it cost	how it was measured
Attention	scanned area of the image **11.1** → **9.5%**; missed cancers **29** → **61** out of a hundred	eye-tracking study; subgroup of 14 cases
Skill	findings **284** → **224** per thousand procedures, same nineteen doctors	observational, measured once, and disputed
Signature	**9 of 11** machine-written discharge summaries contained an error	one hundred hospitalizations, two blind readers
Arbitration	readings sent to arbitration up **threefold and by a quarter**; one review takes **as long as five readings**	two centers, 50,000 women; 441,276 cases

Paid unevenly, and the order of magnitude matters more than the figure: arbitration has two trials and the evening's largest samples; attention has a subgroup of 14 cases; skill has one unreplicated observation; signature has a share of documents, not an outcome.

Who pays: screening — in arbitration · diagnosis — in attention · documents — in signature · qualification — in skill. **Instrumental diagnostics pays none of the four:** its cost is 290 missed findings, a different plane entirely.

All six distortions from the front door showed up: anchoring and narrowing — in attention, decay — in skill, correction — in signature, disagreement — in arbitration; alert stream got no column of its own. **Exactly one is named by its own name over the evening.**

What the lecture used to measure the person's work

We measured the person's work, not what happened to the patient

screening is the one that reached the patient's outcome: 1.55 interval cancers against 1.76 per thousand — showing non-inferiority, not improvement

MASAI final analysis, The Lancet 2026

The four rungs of human participation, and the three grounds for their signature

① LOOKS FIRST

reads it themselves, the model is a second opinion

disagreement goes to arbitration · randomized, 105,934 women

② LOOKS SECOND

reads after the model

attention narrows, a correct call flips to incorrect · eye-tracking study and review

③ LOOKS AT DISPUTED CASES

reviews what's flagged, the rest goes untouched

55 scans out of a hundred are read by no one

④ SIGNS WITHOUT LOOKING

1.99 seconds

median time checking the suggestion; measured outside medicine

Human participation is required, and not because they're more accurate — it's been measured that they often aren't. The question is what they do after the model, and who measured that.

The lower the rung, the weaker the evidence. The fourth rung, also the most common, is the one we know the least about: the signature is there, the reading is not.

All four showed up over one case — the woman called in by letter by her year of birth: ① at screening, ② on her scan, ③ on her preventive chest X-ray, ④ on her discharge.

Three grounds, and they hold up differently. *The right to sign:* a case review evaluates the person's actions, and there's no question of the model in that procedure at all. *Errs differently:* across 106 studies, the sign of the pair's payoff depends on which side is stronger — the ground has no magnitude. *Cannot be undone:* a surgery can't be undone after the fact, a year of follow-up can't be replayed.

The rule is phrased as a capacity to intervene — presence does not satisfy it. And not one of the four rungs has been measured through to what the decision did to the patient.

Tacconelli et al., J Nucl Med 2026 · *EU Regulation 2024/1689, Article 14*

The device gets recalled; the decisions stay

WHAT GETS RECALLED

the device

the recall procedure exists — the regulator's record sits alongside

WHAT STAYS

the decisions made during its year in service

a procedure can't be redone, a year of follow-up can't be replayed



"In medicine everything is irreversible" is a claim wider than any measurement. **Devices get recalled: the regulator's record sits right here.** Decisions made on a device during its year in service don't get recalled along with it.

What happened to the woman with the letter, nobody measured. The same question for your own loop: what in it gets recalled, and what stays.

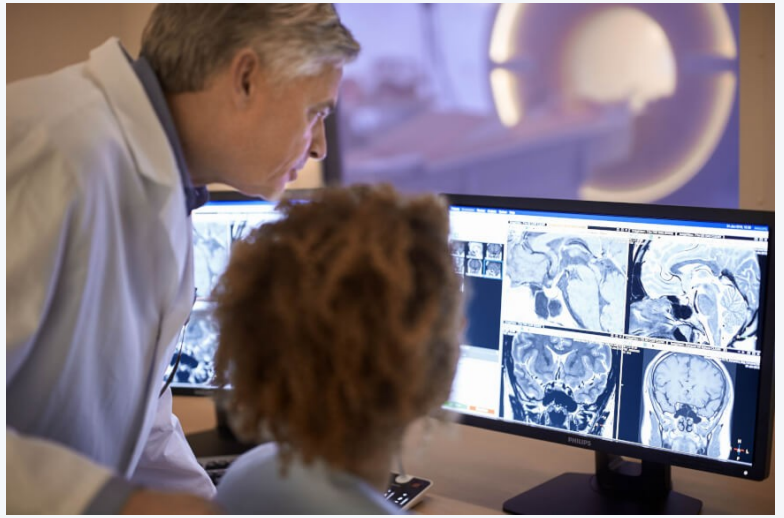
Next — an industry where an error can almost always be undone: the dispute there is over who owns the result; the cost of error barely gets a mention.

Lee et al., JAMA Health Forum 2025, cutoff November 2024 · FDA device recall database, record Z-2111-2021, October 2026

Three checks for any industry — and a question

Is there an independent source of truth besides the model's own explanation? A different instrument, a different analysis, a different person, or the thing itself. A check against self-explanation is not a check.

Has it been measured that they object — and that they object correctly? Two countable rates. Neither is measured — "the person checks" is no different from "the person signs."



Can they physically keep up? Seconds for the decision against the volume to read. The median time checking a suggestion is the same **1.99 seconds** that came up on the evening's axis; what's accepted without a check gets corrected afterward in **53% of cases against 18**.

If you were putting a model into your own loop tomorrow — which one check would you make mandatory before switching it on?